Exemple d'utilisation du package DegradSim

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Préparation

Installation du package depuis GitHub :

```
remotes::install_github("rytracetito/levy_packages", subdir = "DegradSim")
```

Les parties I et II utilisent les **processus de Lévy** (cas homogène, `A = NULL` par défaut). La partie III montre les **processus de Lévy non homogènes** $X_t = Y_{A(t)}$ obtenus par un changement de temps A : gamma, Wiener et inverse gaussien.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 0.5`, `sigma = 1.5`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`

Dans chaque cas, les trajectoires simulées sont d'abord **stockées** dans un objet `x`, puis utilisées : premières valeurs, puis graphique. Quand toutes les trajectoires ont les mêmes instants d'inspection, `x` est une matrice :

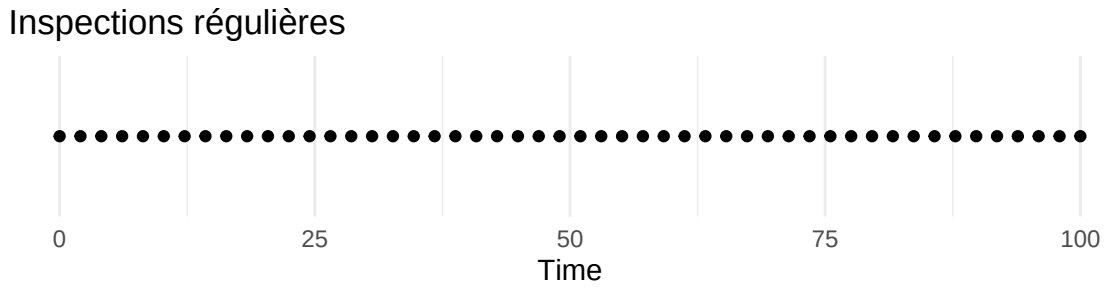
une colonne `t`, puis une colonne par trajectoire (`X1`, `X2`, ...).

I - Une trajectoire

Une trajectoire inspectée 50 fois entre 0 et 100. `x` est une matrice à deux colonnes : `t` et `X1`.

Cas 1 : inspections régulières

```
t1 <- inspection_times("regular", n = 49, tmax = 100)$time
plot_plan(t1, "Inspections régulières")
```

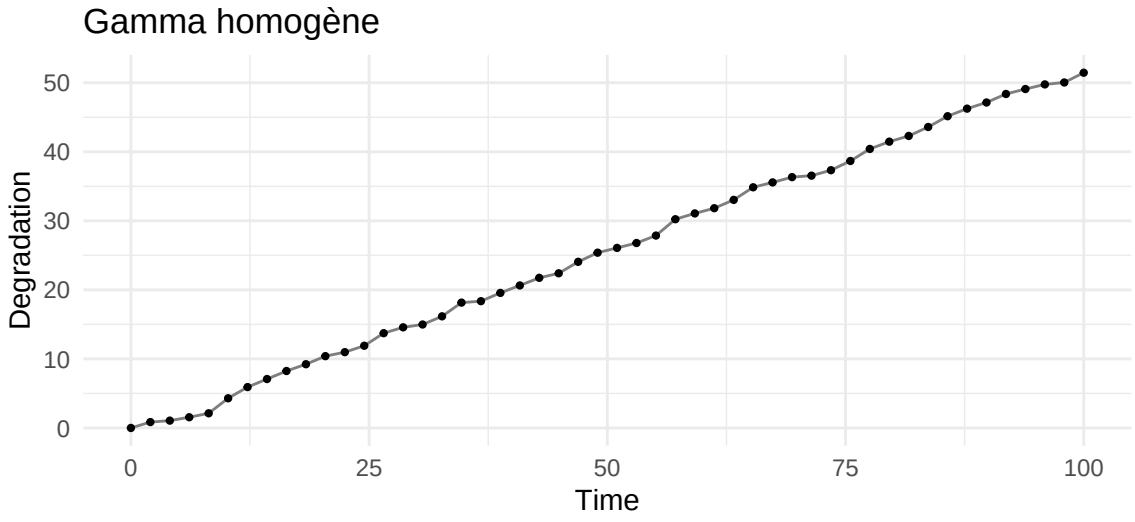


Modèle 1 : gamma

```
set.seed(11)
x <- sim_gamma(t1, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.040816 0.8501413
## [3,]  4.081633 1.0605764
## [4,]  6.122449 1.5495729
## [5,]  8.163265 2.1242214
## [6,] 10.204082 4.2990767
```

```
plot_traj(x, "Gamma homogène")
```

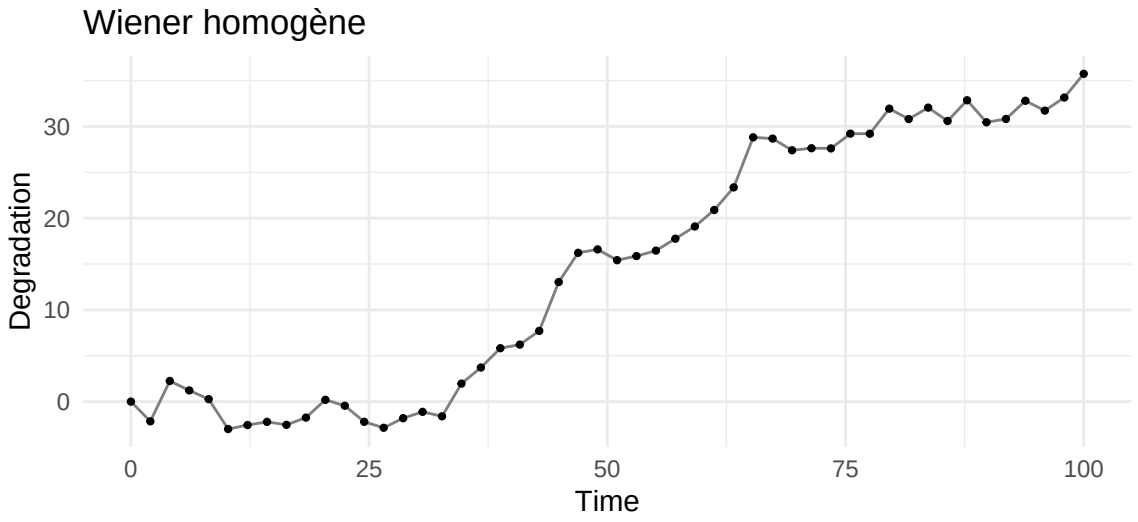


Modèle 2 : Wiener

```
set.seed(12)
x <- sim_wiener(t1, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t         X1
## [1,]  0.000000  0.0000000
## [2,]  2.040816 -2.1522367
## [3,]  4.081633  2.2478203
## [4,]  6.122449  1.2180618
## [5,]  8.163265  0.2670301
## [6,] 10.204082 -2.9932234
```

```
plot_traj(x, "Wiener homogène")
```

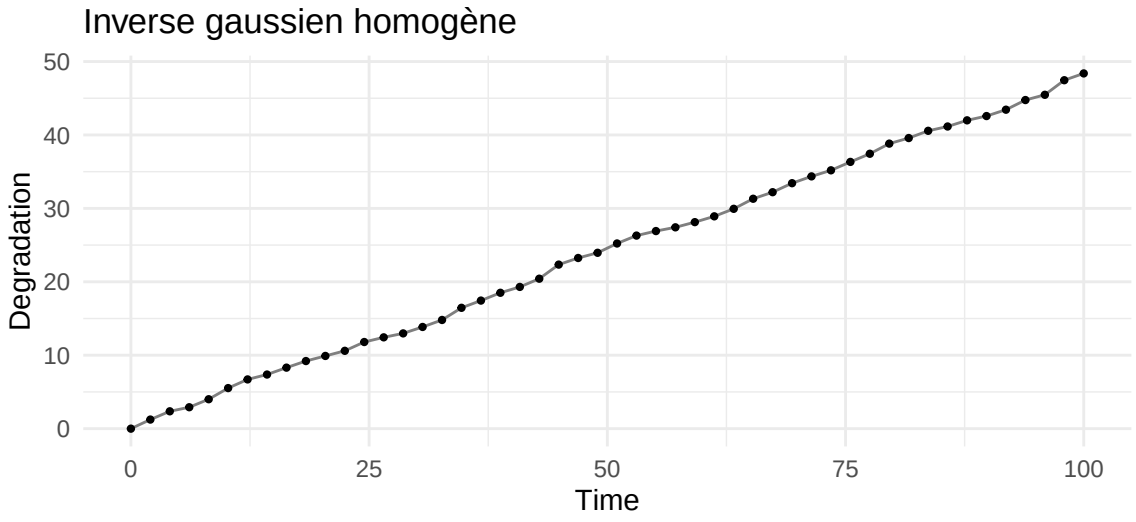


Modèle 3 : inverse gaussien

```
set.seed(13)
x <- sim_ig(t1, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  2.040816 1.238516
## [3,]  4.081633 2.364051
## [4,]  6.122449 2.917536
## [5,]  8.163265 4.007073
## [6,] 10.204082 5.525188
```

```
plot_traj(x, "Inverse gaussien homogène")
```

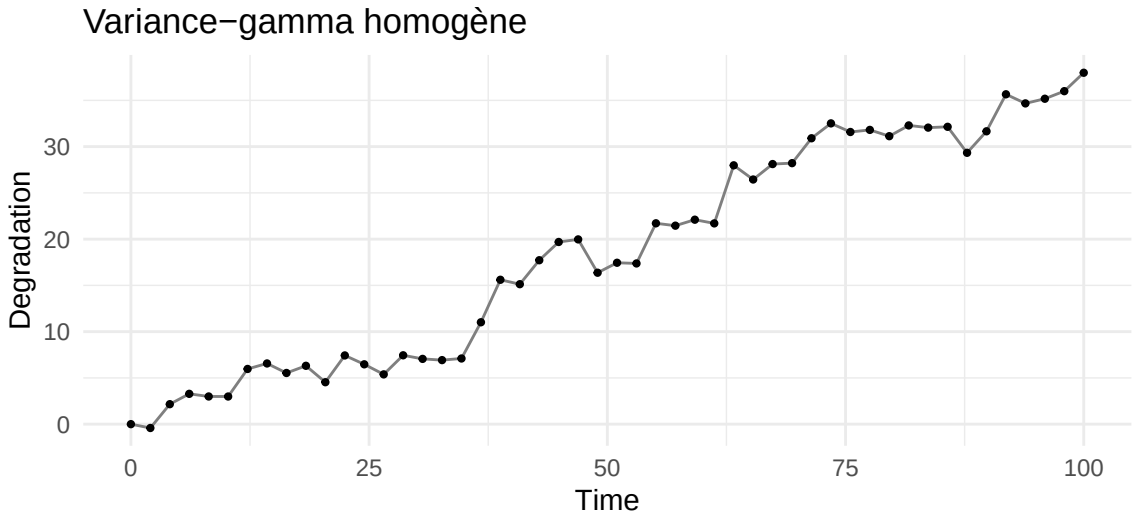


Modèle 4 : variance-gamma

```
set.seed(14)
x <- sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  2.040816 -0.414327
## [3,]  4.081633  2.158973
## [4,]  6.122449  3.277011
## [5,]  8.163265  2.991563
## [6,] 10.204082  2.988429
```

```
plot_traj(x, "Variance-gamma homogène")
```

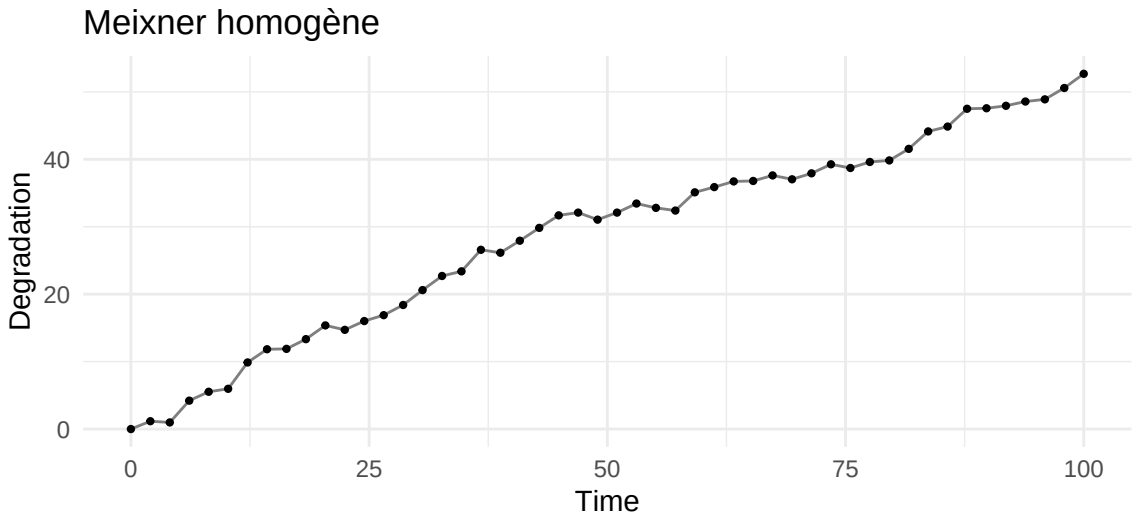


Modèle 5 : Meixner

```
set.seed(15)
x <- sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

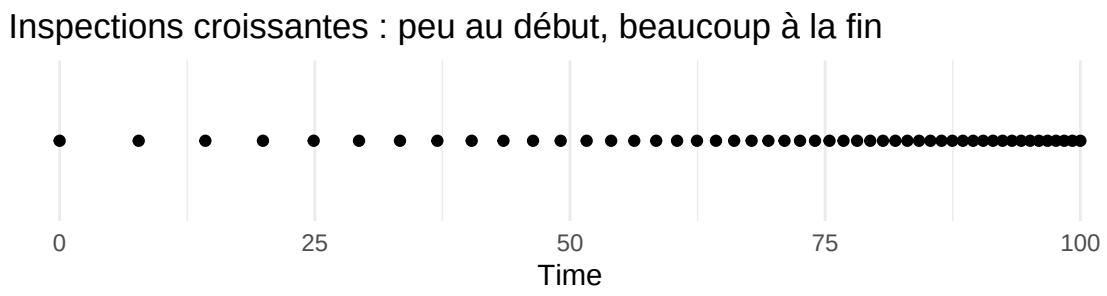
```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  2.040816 1.1594818
## [3,]  4.081633 0.9808625
## [4,]  6.122449 4.2100728
## [5,]  8.163265 5.5272998
## [6,] 10.204082 5.9679779
```

```
plot_traj(x, "Meixner homogène")
```



Cas 2 : inspections croissantes

```
t2 <- inspection_times("increasing", n = 49, tmax = 100)$time
plot_plan(t2, "Inspections croissantes : peu au début, beaucoup à la fin")
```

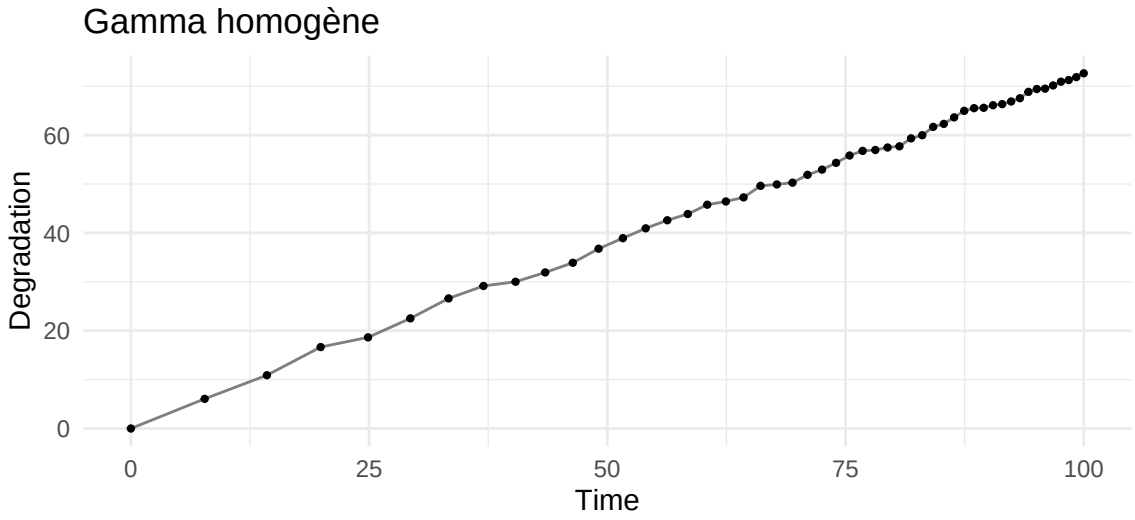


Modèle 1 : gamma

```
set.seed(21)
x <- sim_gamma(t2, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  6.072512
## [3,] 14.274444 10.898459
## [4,] 19.918616 16.660539
## [5,] 24.889163 18.640144
## [6,] 29.329869 22.520703
```

```
plot_traj(x, "Gamma homogène")
```

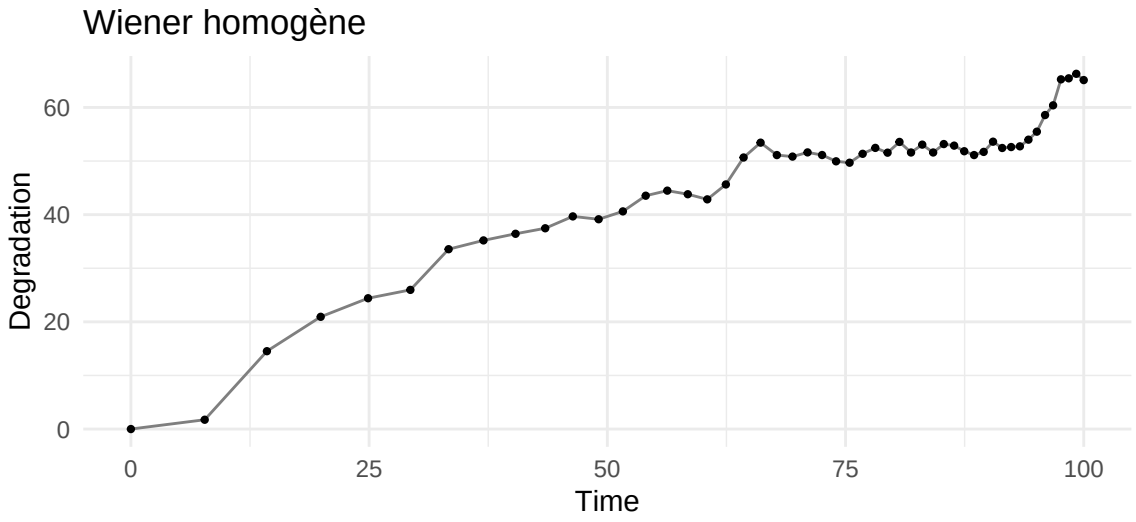


Modèle 2 : Wiener

```
set.seed(22)
x <- sim_wiener(t2, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  7.745007  1.73459
## [3,] 14.274444 14.52480
## [4,] 19.918616 20.93840
## [5,] 24.889163 24.40291
## [6,] 29.329869 25.96275
```

```
plot_traj(x, "Wiener homogène")
```

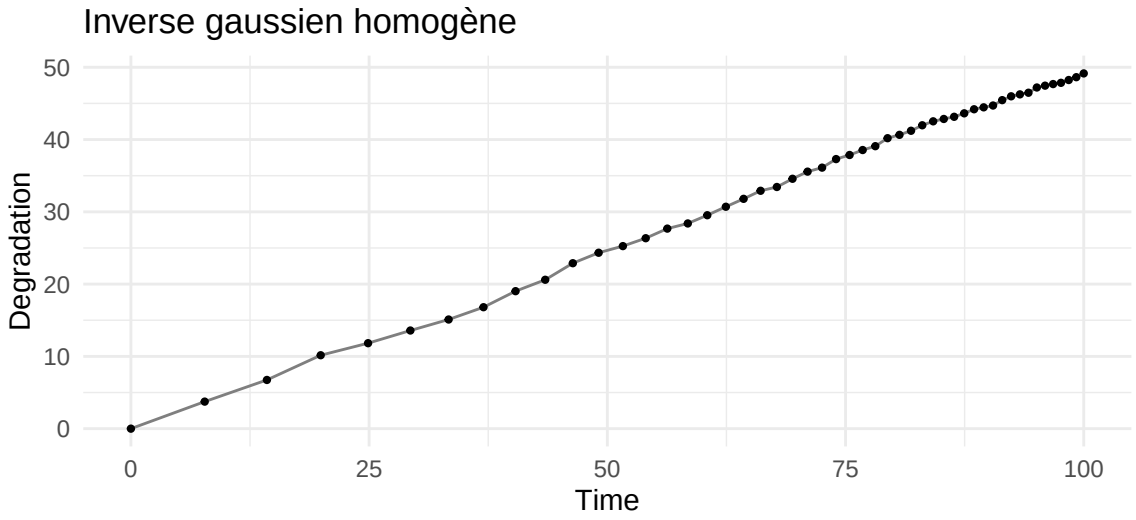


Modèle 3 : inverse gaussien

```
set.seed(23)
x <- sim_ig(t2, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  3.740389
## [3,] 14.274444  6.738983
## [4,] 19.918616 10.158120
## [5,] 24.889163 11.824819
## [6,] 29.329869 13.578556
```

```r
plot_traj(x, "Inverse gaussien homogène")
```

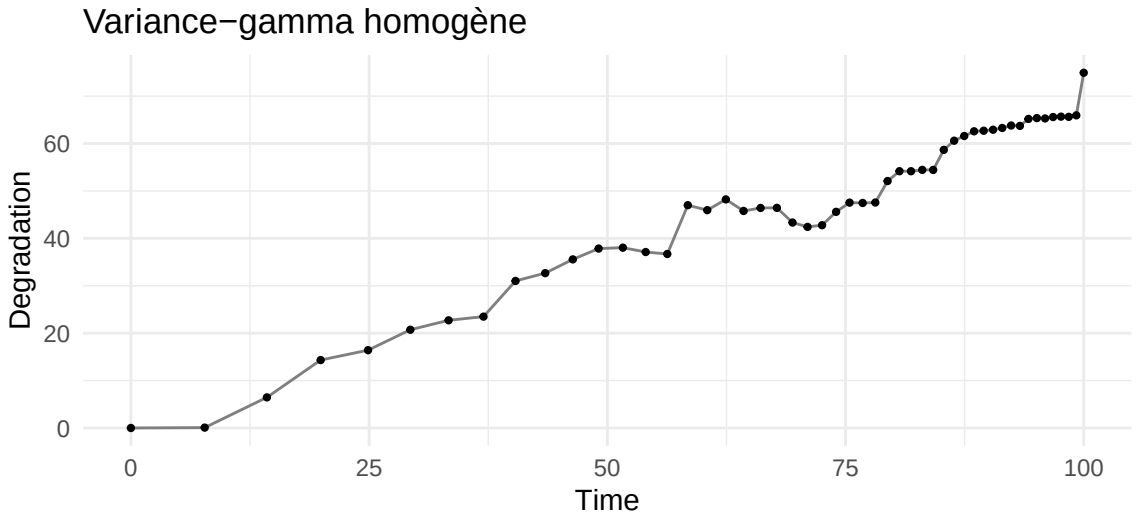


Modèle 4 : variance-gamma

```r
set.seed(24)
x <- sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,]  0.000000  0.00000000
## [2,]  7.745007  0.07862292
## [3,] 14.274444  6.45985339
## [4,] 19.918616 14.32829297
## [5,] 24.889163 16.41061317
## [6,] 29.329869 20.72743266
```

```r
plot_traj(x, "Variance-gamma homogène")
```

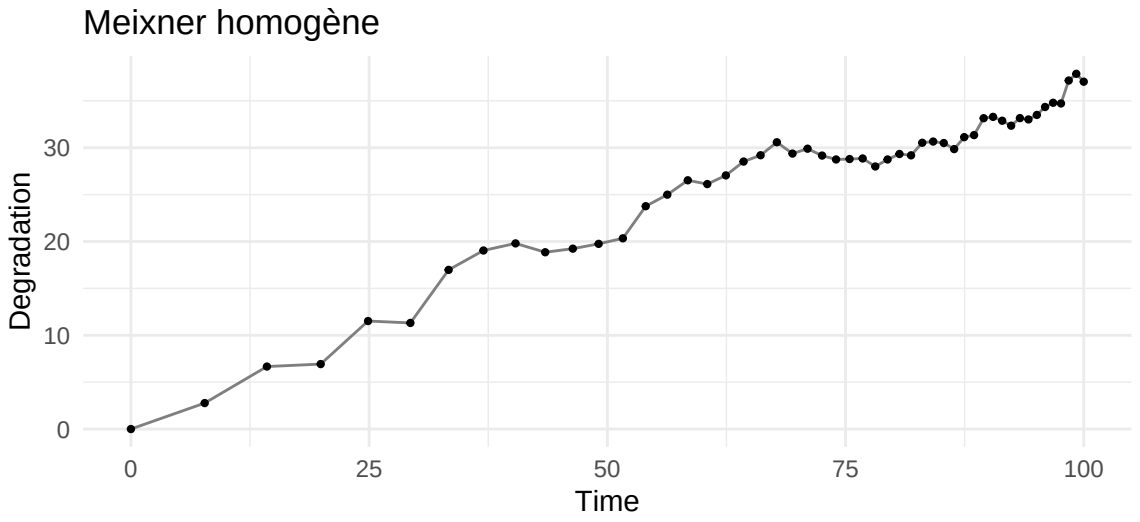


Modèle 5 : Meixner

```
set.seed(25)
x <- sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

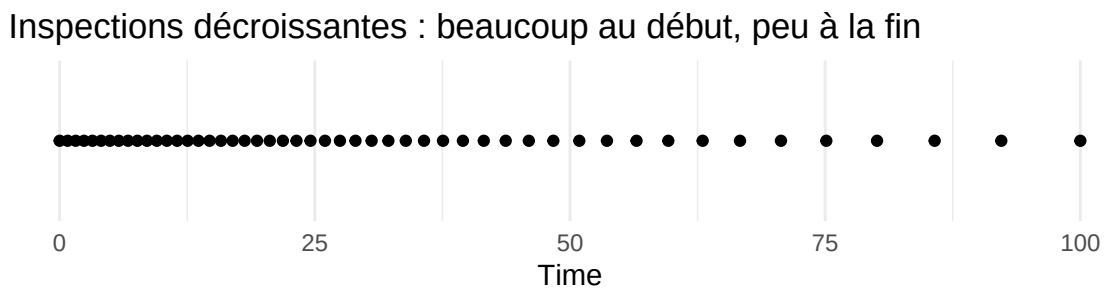
```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  7.745007  2.770050
## [3,] 14.274444  6.664740
## [4,] 19.918616  6.928232
## [5,] 24.889163 11.519308
## [6,] 29.329869 11.315784
```

```
plot_traj(x, "Meixner homogène")
```



Cas 3 : inspections décroissantes

```
t3 <- inspection_times("decreasing", n = 49, tmax = 100)$time
plot_plan(t3, "Inspections décroissantes : beaucoup au début, peu à la fin")
```

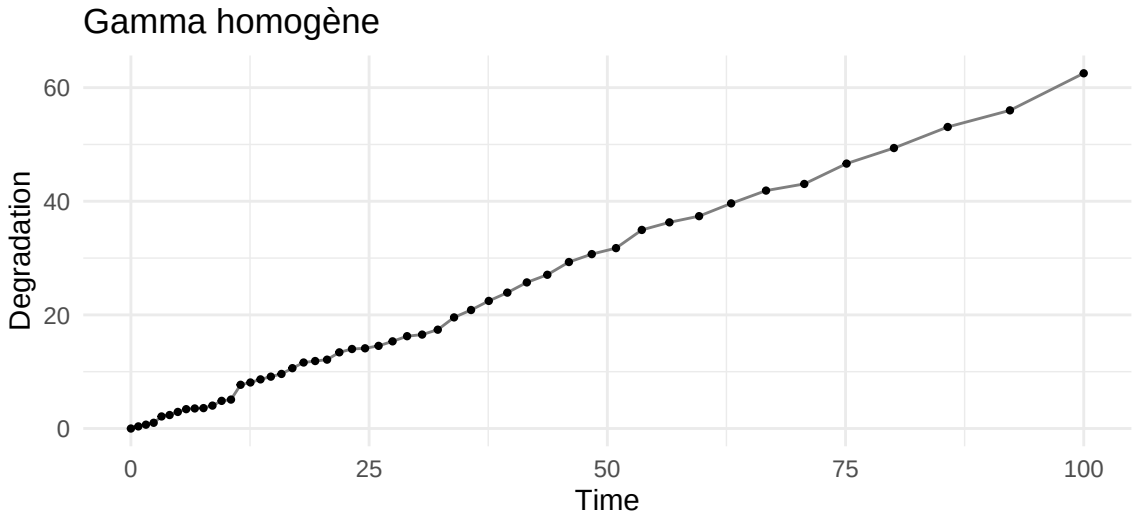


Modèle 1 : gamma

```
set.seed(31)
x <- sim_gamma(t3, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.3733343
## [3,] 1.5768699 0.6759188
## [4,] 2.3882425 1.0089026
## [5,] 3.2157149 2.1225410
## [6,] 4.0599388 2.3735653
```

```
plot_traj(x, "Gamma homogène")
```

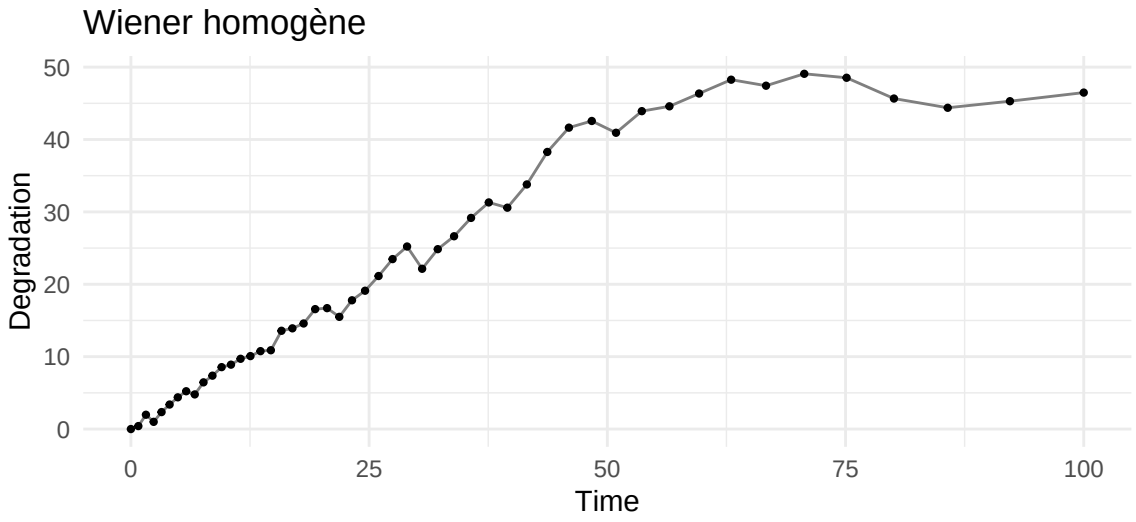


Modèle 2 : Wiener

```
set.seed(32)
x <- sim_wiener(t3, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.4098987
## [3,] 1.5768699 1.9764668
## [4,] 2.3882425 0.9932504
## [5,] 3.2157149 2.3425640
## [6,] 4.0599388 3.3841013
```

```
plot_traj(x, "Wiener homogène")
```

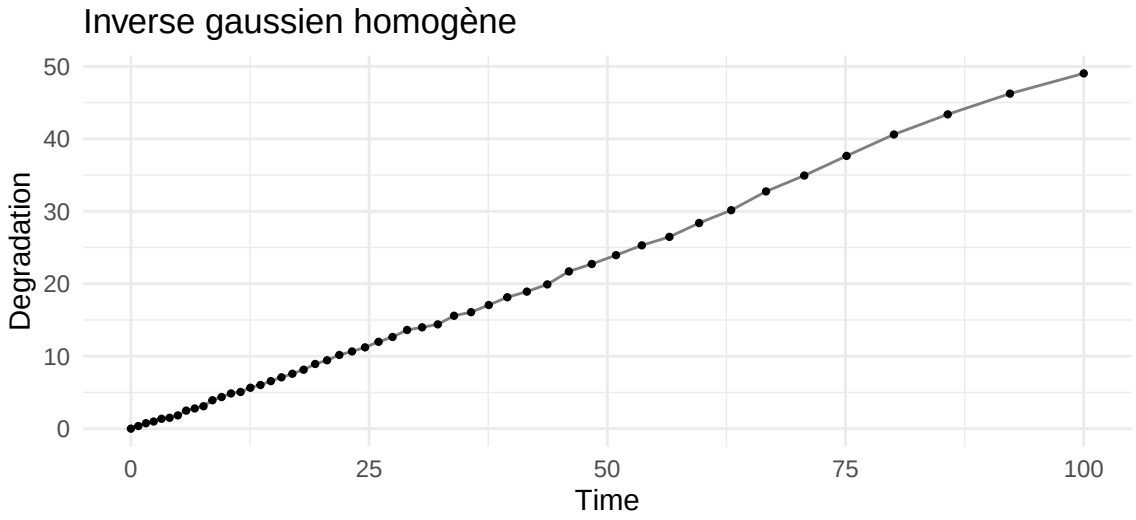


Modèle 3 : inverse gaussien

```
set.seed(33)
x <- sim_ig(t3, mu = 0.5, eta = 2)
head(x)
```

```
##              t        X1
## [1,] 0.0000000 0.0000000
## [2,] 0.7809824 0.3615935
## [3,] 1.5768699 0.7505417
## [4,] 2.3882425 0.9837075
## [5,] 3.2157149 1.3629841
## [6,] 4.0599388 1.5014091
```

```r
plot_traj(x, "Inverse gaussien homogène")
```

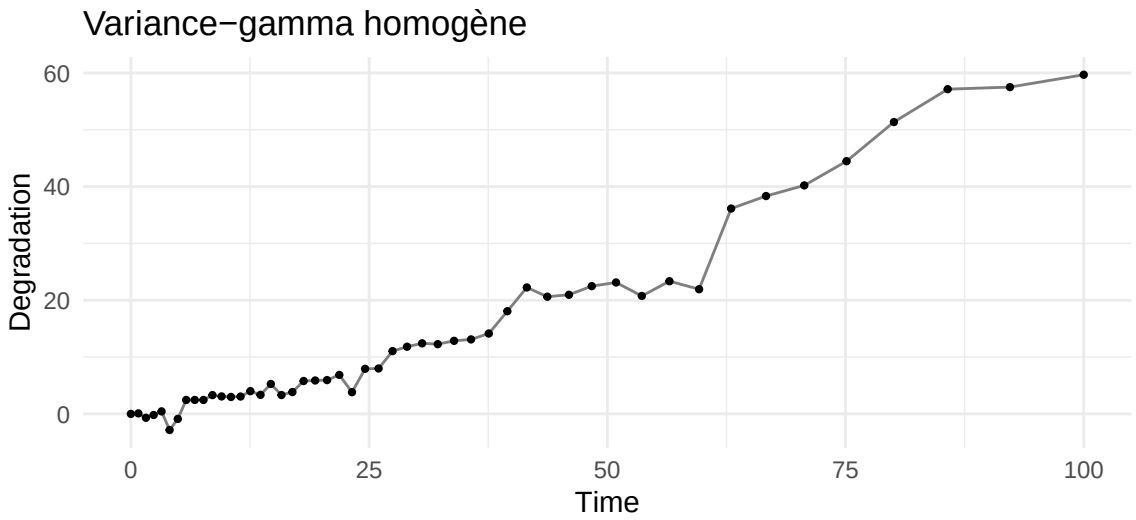


Modèle 4 : variance-gamma

```r
set.seed(34)
x <- sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t          X1
## [1,] 0.0000000  0.00000000
## [2,] 0.7809824  0.09322591
## [3,] 1.5768699 -0.68209708
## [4,] 2.3882425 -0.18811757
## [5,] 3.2157149  0.44419564
## [6,] 4.0599388 -2.82976129
```

```r
plot_traj(x, "Variance-gamma homogène")
```

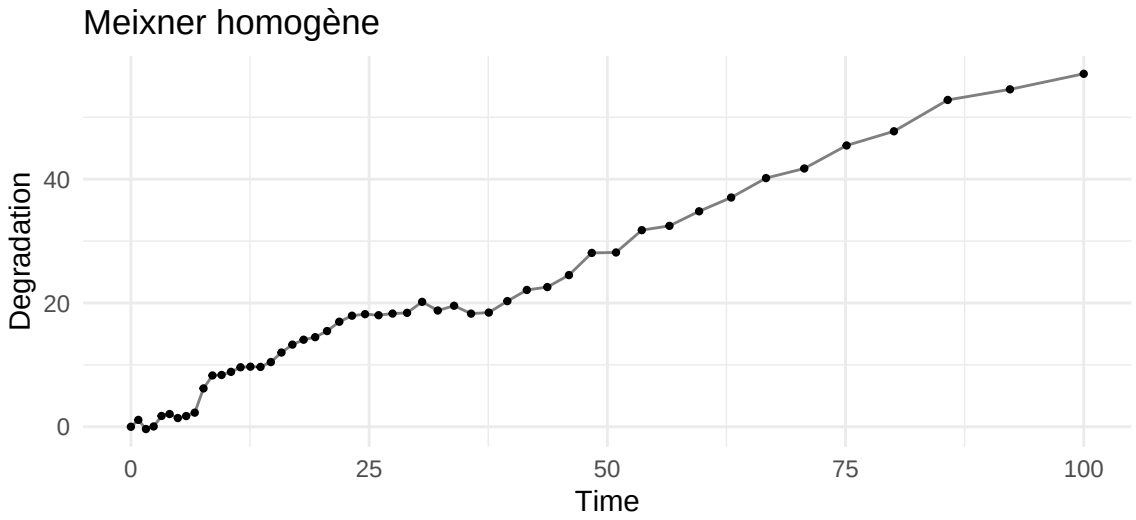


Modèle 5 : Meixner

```
set.seed(35)
x <- sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t          X1
## [1,] 0.0000000  0.00000000
## [2,] 0.7809824  1.11330276
## [3,] 1.5768699 -0.36680705
## [4,] 2.3882425  0.05187965
## [5,] 3.2157149  1.74654076
## [6,] 4.0599388  2.05897099
```

```
plot_traj(x, "Meixner homogène")
```



Cas 4 : mixtes, beaucoup au début et à la fin

```
t4 <- inspection_times("dec-inc", n = 49, tmax = 100)$time
plot_plan(t4, "Beaucoup au début et à la fin, peu au milieu")
```

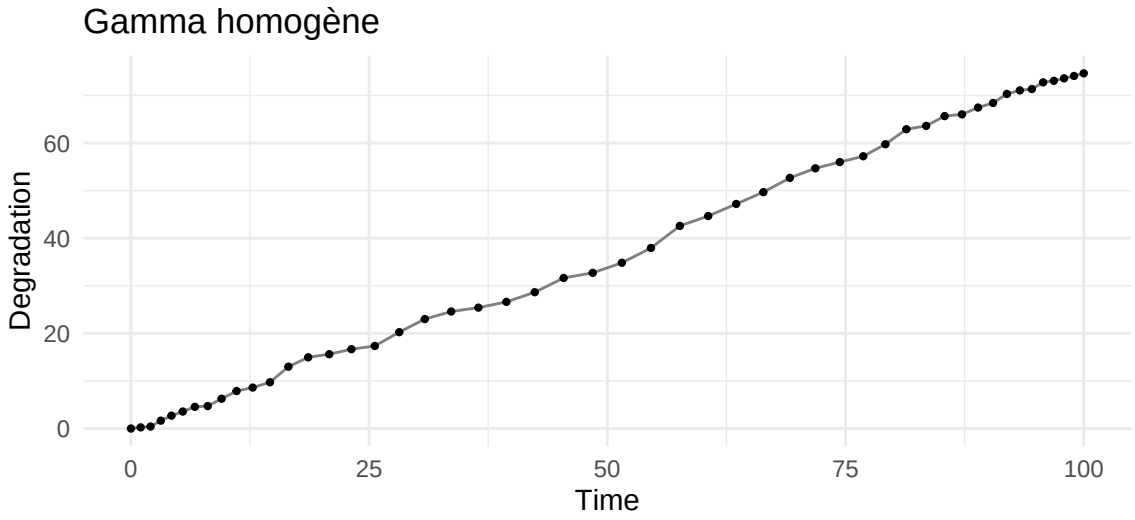


Modèle 1 : gamma

```
set.seed(41)
x <- sim_gamma(t4, a = 2, b = 3)
head(x)
```

```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.2387756
## [3,] 2.063114 0.4122480
## [4,] 3.136169 1.6505655
## [5,] 4.258259 2.6990597
## [6,] 5.444469 3.5686811
```

```
plot_traj(x, "Gamma homogène")
```

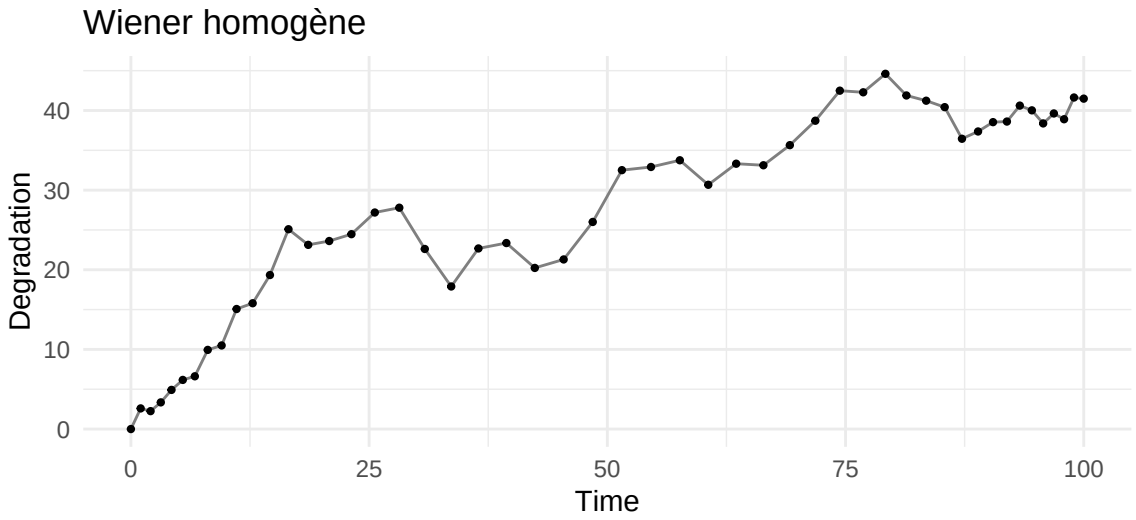


Modèle 2 : Wiener

```
set.seed(42)
x <- sim_wiener(t4, mu = 0.5, sigma = 1.5)
head(x)
```

```
##             t       X1
## [1,] 0.000000 0.000000
## [2,] 1.023202 2.591759
## [3,] 2.063114 2.247929
## [4,] 3.136169 3.348695
## [5,] 4.258259 4.915316
## [6,] 5.444469 6.168873
```

```
plot_traj(x, "Wiener homogène")
```

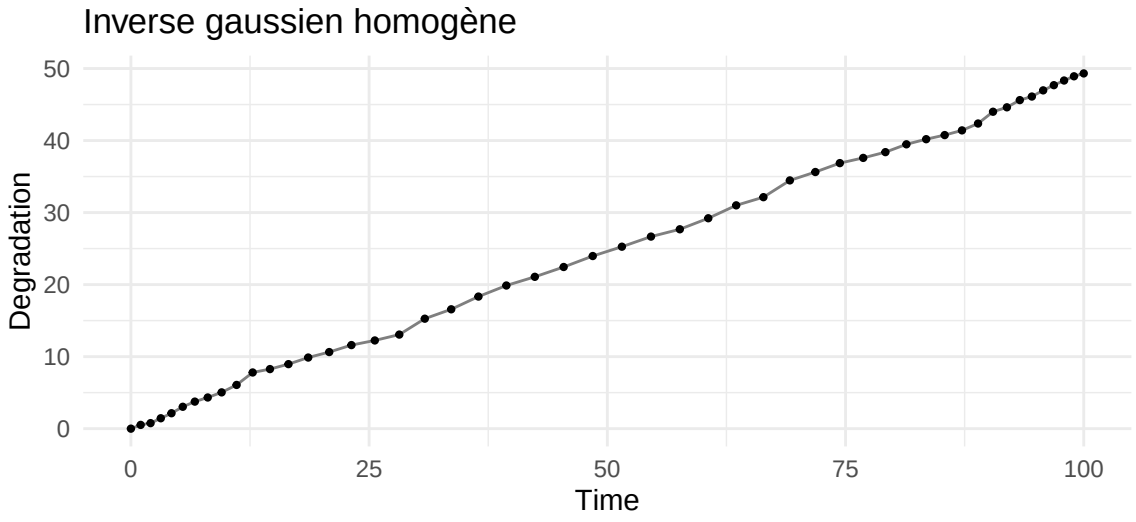


Modèle 3 : inverse gaussien

```
set.seed(43)
x <- sim_ig(t4, mu = 0.5, eta = 2)
head(x)
```

```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.5211761
## [3,] 2.063114 0.7657921
## [4,] 3.136169 1.4437941
## [5,] 4.258259 2.1422958
## [6,] 5.444469 3.0379037
```

```
plot_traj(x, "Inverse gaussien homogène")
```

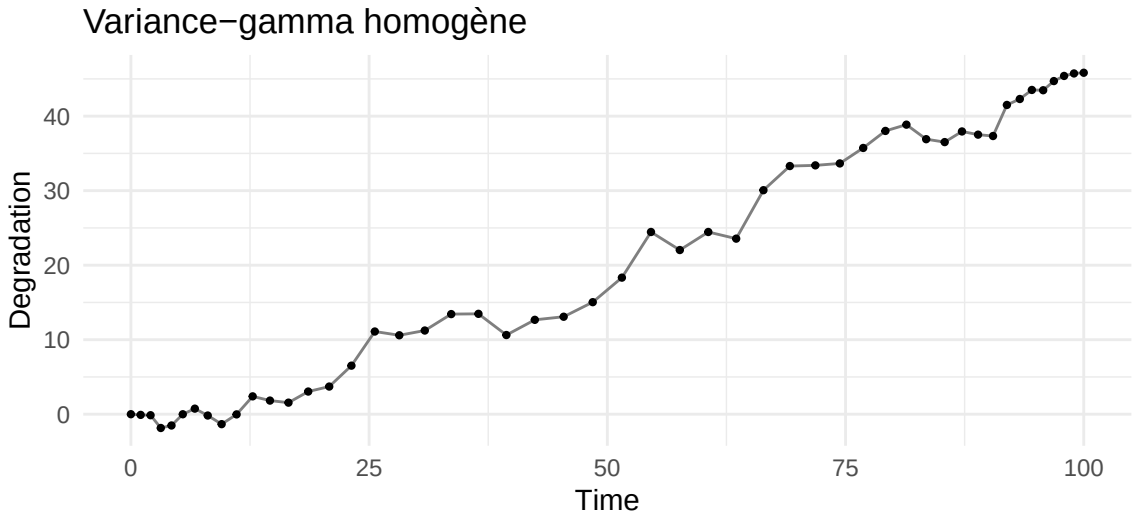


Modèle 4 : variance-gamma

```
set.seed(44)
x <- sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##             t           X1
## [1,] 0.000000  0.000000000
## [2,] 1.023202 -0.091184314
## [3,] 2.063114 -0.136224319
## [4,] 3.136169 -1.849759658
## [5,] 4.258259 -1.513628228
## [6,] 5.444469 -0.007804398
```

```
plot_traj(x, "Variance-gamma homogène")
```

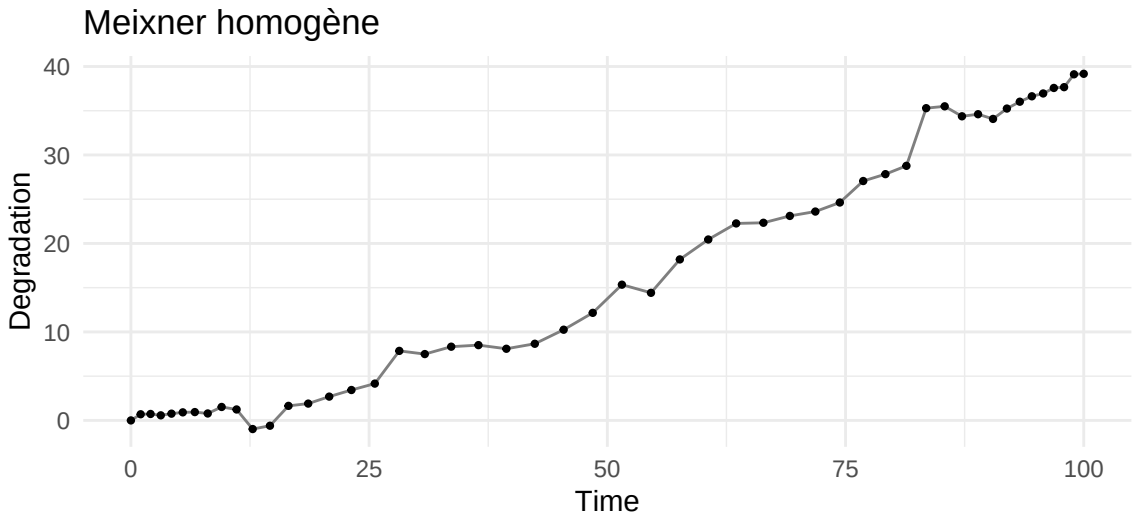


Modèle 5 : Meixner

```
set.seed(45)
x <- sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

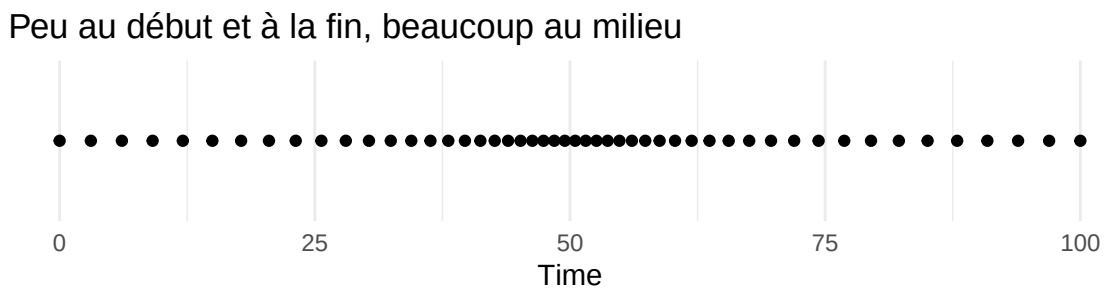
```
##             t        X1
## [1,] 0.000000 0.0000000
## [2,] 1.023202 0.6957971
## [3,] 2.063114 0.7273165
## [4,] 3.136169 0.5743888
## [5,] 4.258259 0.7643220
## [6,] 5.444469 0.9113741
```

```
plot_traj(x, "Meixner homogène")
```



Cas 5 : mixtes, beaucoup au milieu

```
t5 <- inspection_times("inc-dec", n = 49, tmax = 100)$time
plot_plan(t5, "Peu au début et à la fin, beaucoup au milieu")
```

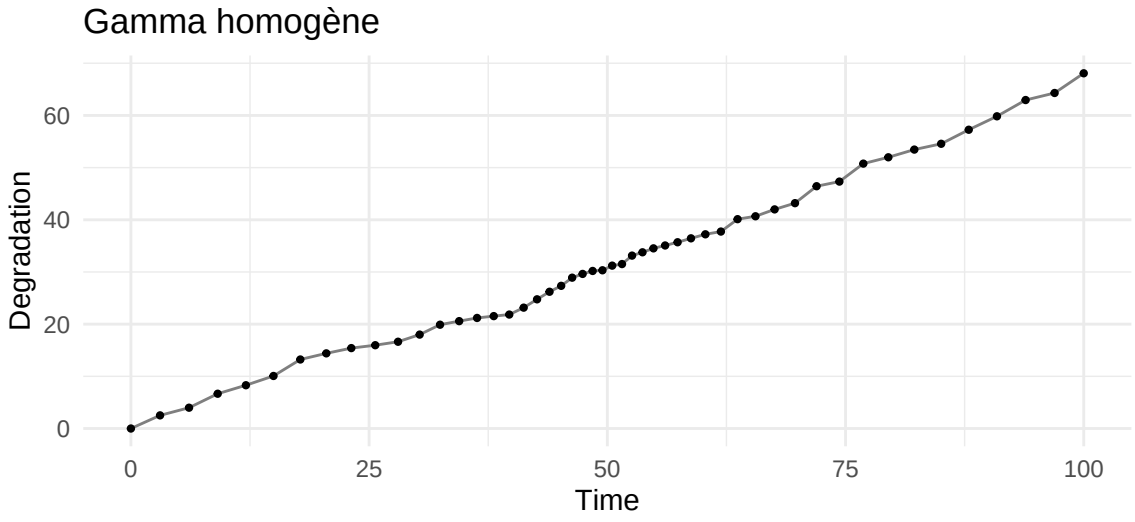


Modèle 1 : gamma

```
set.seed(51)
x <- sim_gamma(t5, a = 2, b = 3)
head(x)
```

```
##              t        X1
## [1,]  0.000000  0.000000
## [2,]  3.058430  2.519601
## [3,]  6.100152  3.987164
## [4,]  9.108729  6.665784
## [5,] 12.068272  8.292320
## [6,] 14.963694 10.070391
```

```
plot_traj(x, "Gamma homogène")
```

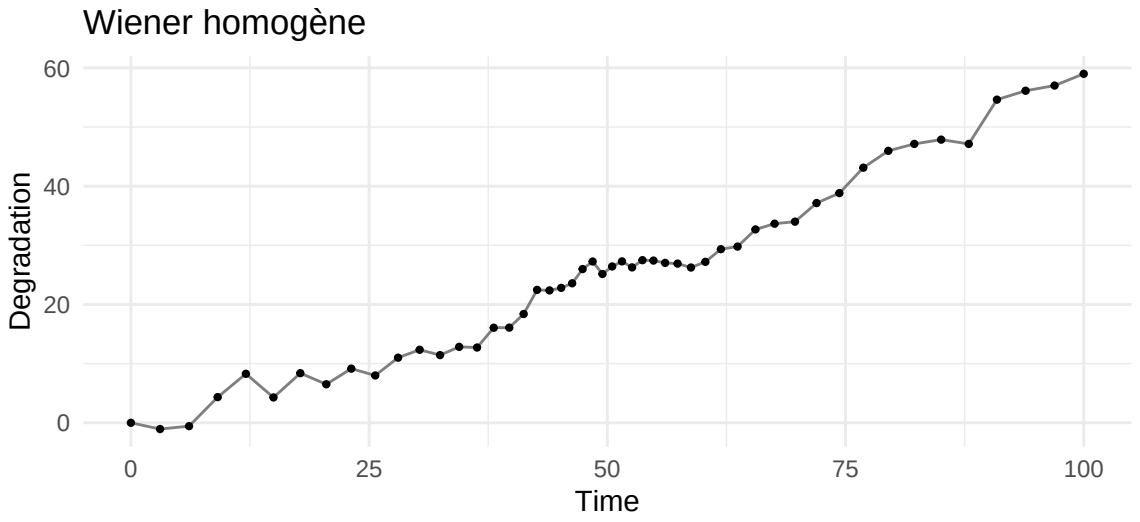


Modèle 2 : Wiener

```
set.seed(52)
x <- sim_wiener(t5, mu = 0.5, sigma = 1.5)
head(x)
```

```
##              t         X1
## [1,]  0.000000  0.0000000
## [2,]  3.058430 -1.0578153
## [3,]  6.100152 -0.5699274
## [4,]  9.108729  4.3388401
## [5,] 12.068272  8.2823679
## [6,] 14.963694  4.2856276
```

```
plot_traj(x, "Wiener homogène")
```

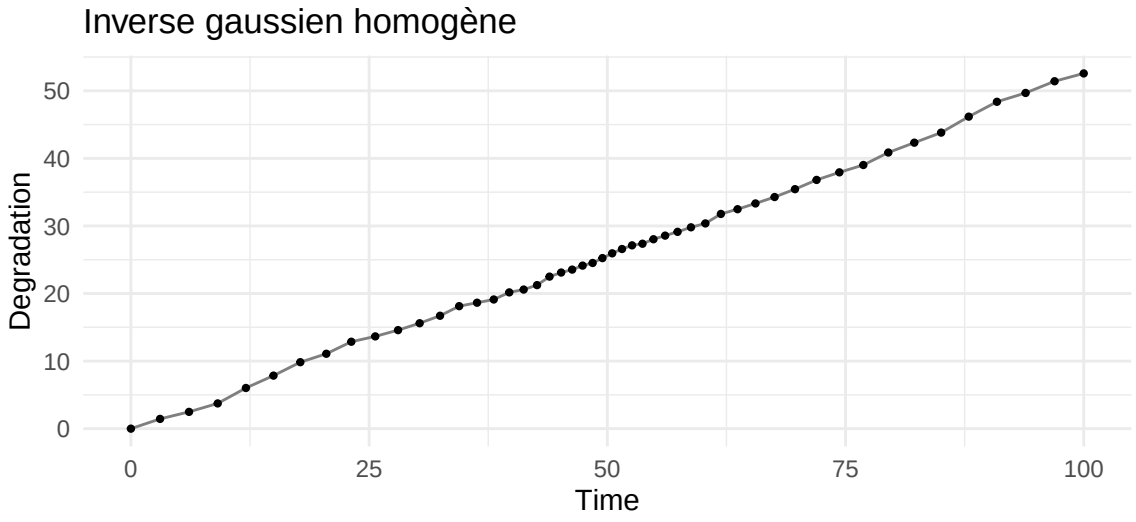


Modèle 3 : inverse gaussien

```
set.seed(53)
x <- sim_ig(t5, mu = 0.5, eta = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000 0.000000
## [2,]  3.058430 1.444313
## [3,]  6.100152 2.483767
## [4,]  9.108729 3.733955
## [5,] 12.068272 6.018209
## [6,] 14.963694 7.847824
```

```r
plot_traj(x, "Inverse gaussien homogène")
```

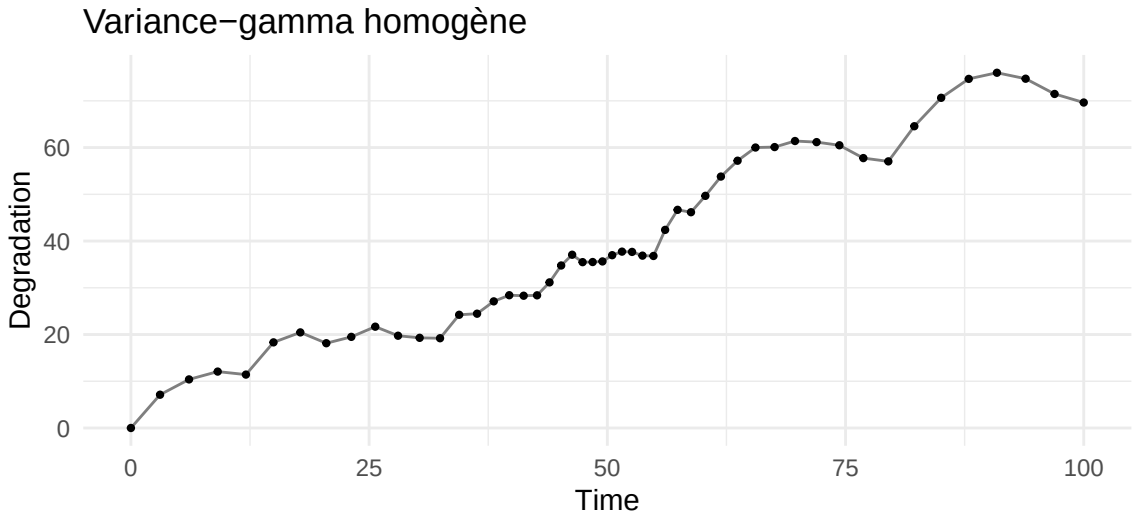


Modèle 4 : variance-gamma

```r
set.seed(54)
x <- sim_vg(t5, theta = 0.5, sigma = 1.5, nu = 2)
head(x)
```

```
##              t       X1
## [1,]  0.000000  0.00000
## [2,]  3.058430  7.11978
## [3,]  6.100152 10.39530
## [4,]  9.108729 12.06922
## [5,] 12.068272 11.42444
## [6,] 14.963694 18.32119
```

```r
plot_traj(x, "Variance-gamma homogène")
```

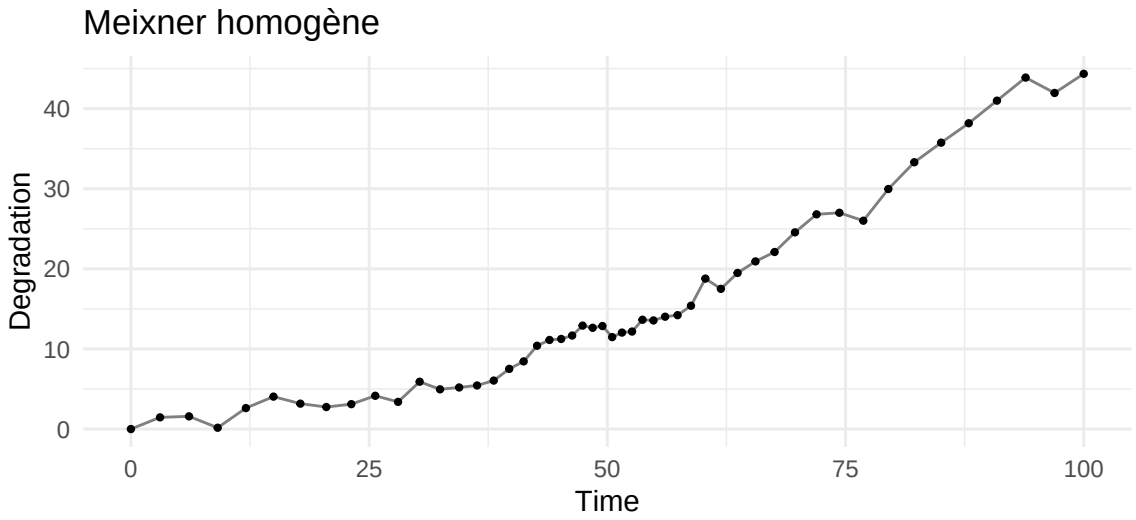


Modèle 5 : Meixner

```
set.seed(55)
x <- sim_meixner(t5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x)
```

```
##              t        X1
## [1,]  0.000000 0.0000000
## [2,]  3.058430 1.4599132
## [3,]  6.100152 1.5847059
## [4,]  9.108729 0.1640323
## [5,] 12.068272 2.6159170
## [6,] 14.963694 4.0469879
```

```
plot_traj(x, "Meixner homogène")
```

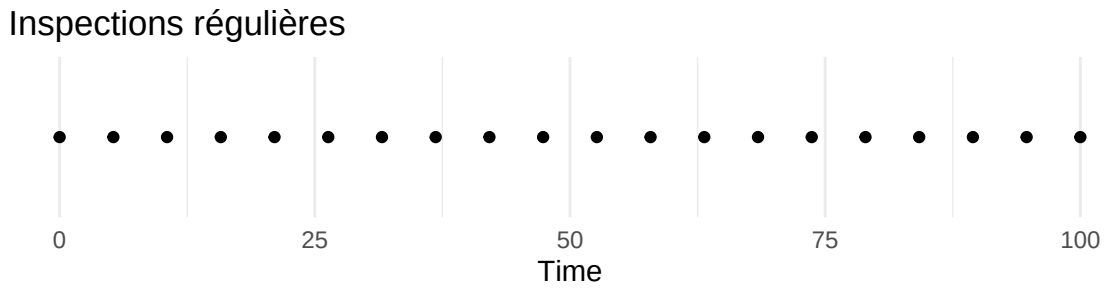


II - Plusieurs trajectoires

10 trajectoires, chacune inspectée 20 fois entre 0 et 100, aux mêmes instants. `x` est une matrice : une colonne `t`, puis `X1`, ..., `X10`. On affiche les premières lignes des trois premières trajectoires.

Cas 1 : inspections régulières

```
p1 <- inspection_times("regular", n = 19, n_traj = 10, tmax = 100)
plot_plan(p1$time[p1$id == 1], "Inspections régulières")
```

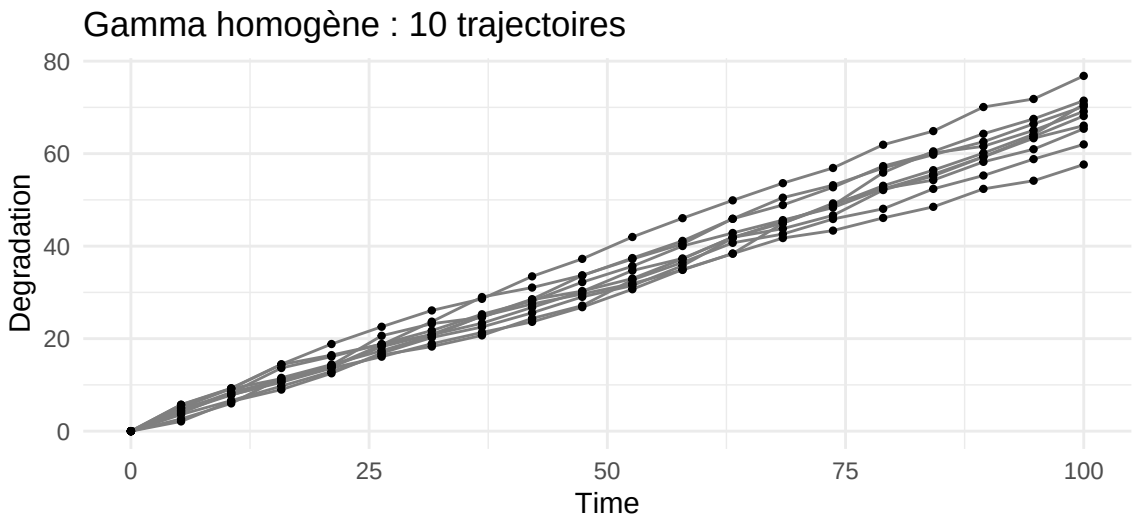


Modèle 1 : gamma

```
set.seed(111)
x <- sim_gamma(p1, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  3.594986  4.263145  5.730951
## [3,] 10.526316  6.597123  7.835829  9.278414
## [4,] 15.789474  9.725184 10.737855 14.460063
## [5,] 21.052632 12.889366 14.284863 18.836785
## [6,] 26.315789 16.476598 20.618768 22.568129
```

```
plot_traj(x, "Gamma homogène : 10 trajectoires")
```

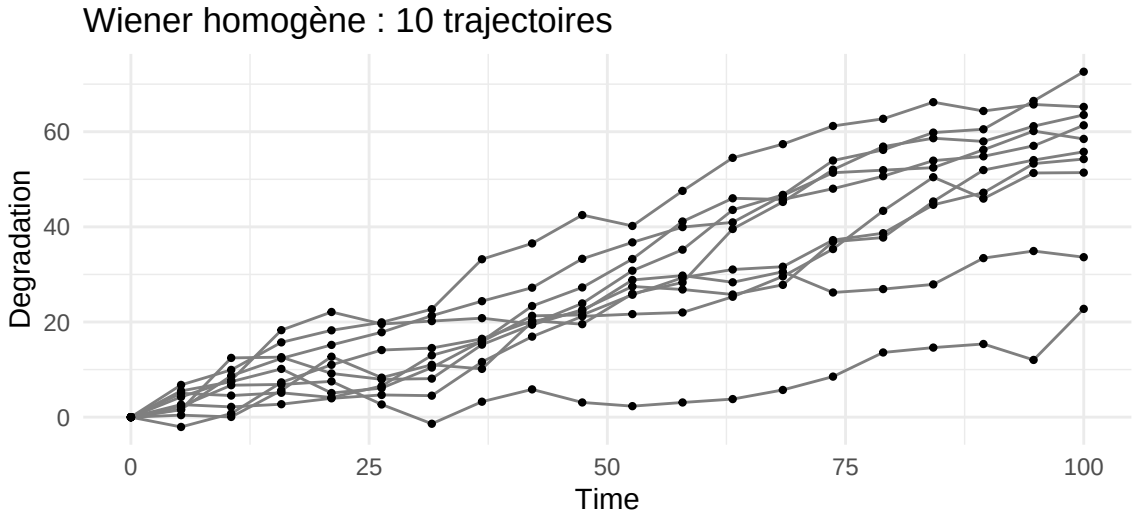


Modèle 2 : Wiener

```
set.seed(112)
x <- sim_wiener(p1, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  1.550234  4.253663  6.772753
## [3,] 10.526316 12.452393  7.741697  9.929144
## [4,] 15.789474 12.612256 18.289009 15.721810
## [5,] 21.052632  9.185158 22.104626 18.251965
## [6,] 26.315789  7.944378 19.582038 19.929558
```

```
plot_traj(x, "Wiener homogène : 10 trajectoires")
```

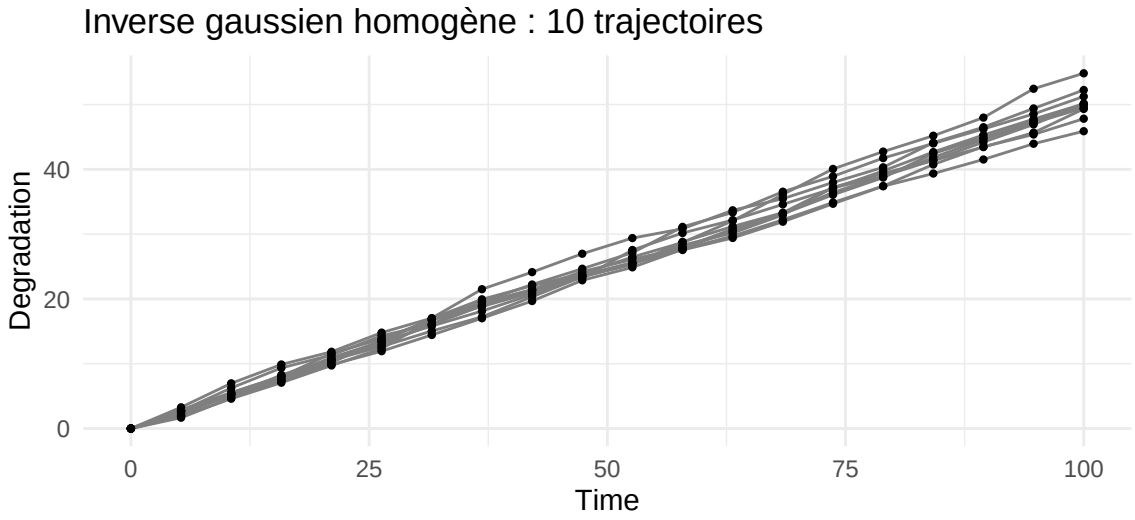


Modèle 3 : inverse gaussien

```
set.seed(113)
x <- sim_ig(p1, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  5.263158  2.709183  2.353263  3.271314
## [3,] 10.526316  6.256515  5.193081  6.976841
## [4,] 15.789474  9.353974  7.519534  9.883285
## [5,] 21.052632 11.340768 10.560528 11.844324
## [6,] 26.315789 14.312931 12.650435 14.798366
```

```
plot_traj(x, "Inverse gaussien homogène : 10 trajectoires")
```

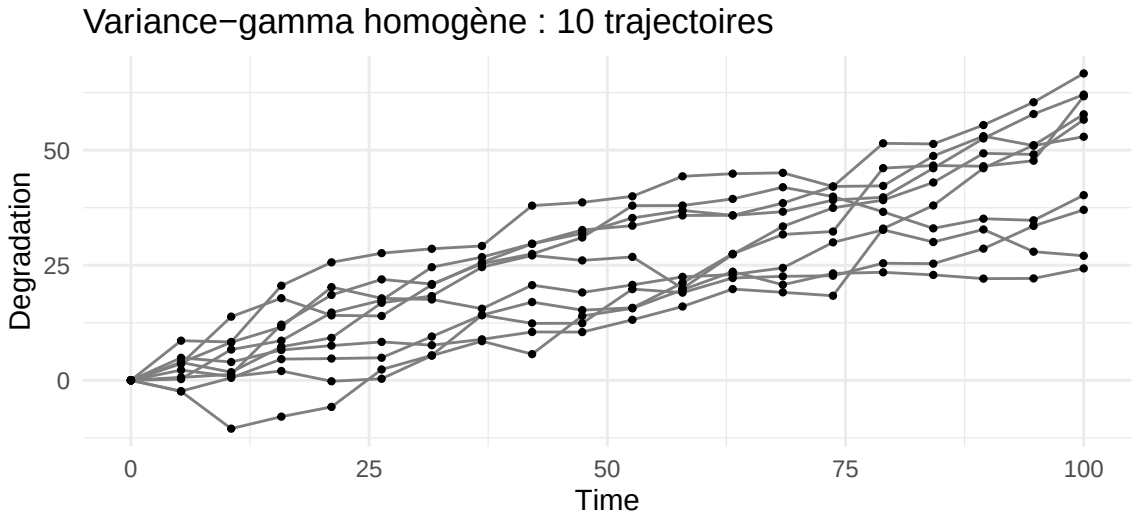


Modèle 4 : variance-gamma

```
set.seed(114)
x <- sim_vg(p1, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1         X2         X3
## [1,]  0.000000  0.000000   0.000000  0.0000000
## [2,]  5.263158  3.823209  -2.413243  2.2232642
## [3,] 10.526316  8.259109 -10.504474  0.8303106
## [4,] 15.789474 11.577788  -7.898515  2.0181671
## [5,] 21.052632 20.242764  -5.784190 -0.2135451
## [6,] 26.315789 17.803908   2.350151  0.3350213
```

```
plot_traj(x, "Variance-gamma homogène : 10 trajectoires")
```

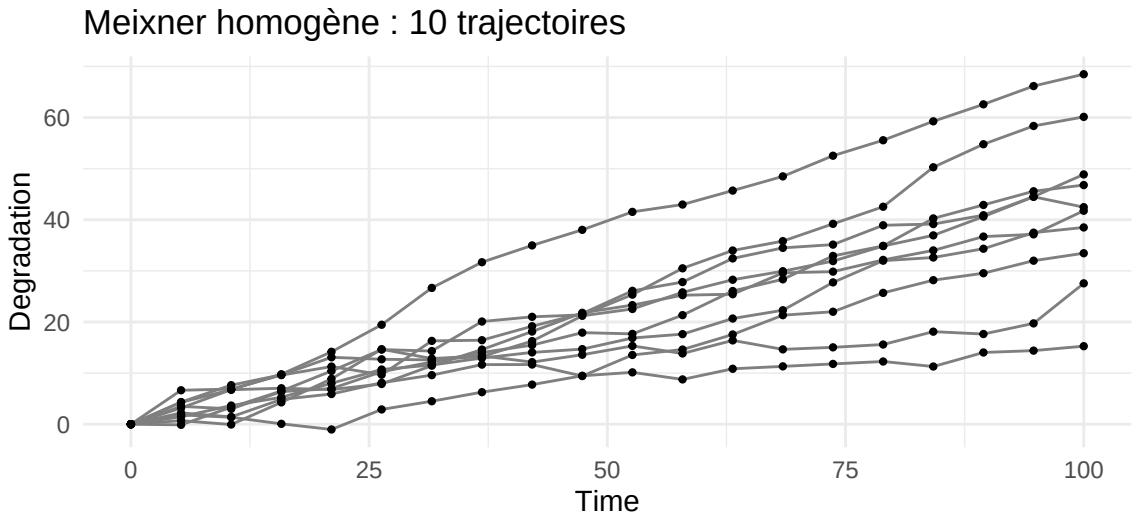


Modèle 5 : Meixner

```
set.seed(115)
x <- sim_meixner(p1, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

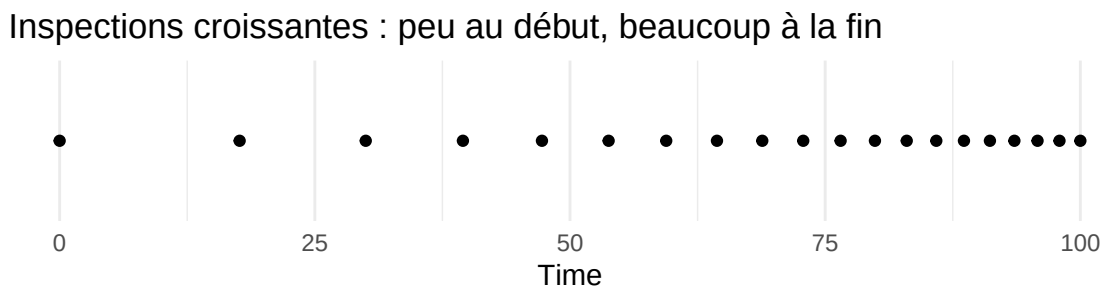
```
##              t        X1        X2       X3
## [1,]  0.000000  0.000000  0.000000 0.000000
## [2,]  5.263158  3.484525  1.464709 4.285754
## [3,] 10.526316  3.010297  3.654886 6.750845
## [4,] 15.789474  6.373638  5.254512 7.040562
## [5,] 21.052632  6.969703  8.015926 6.776480
## [6,] 26.315789 10.268732 10.705041 7.873784
```

```
plot_traj(x, "Meixner homogène : 10 trajectoires")
```



Cas 2 : inspections croissantes

```
p2 <- inspection_times("increasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p2$time[p2$id == 1], "Inspections croissantes : peu au début, beaucoup à la fin")
```

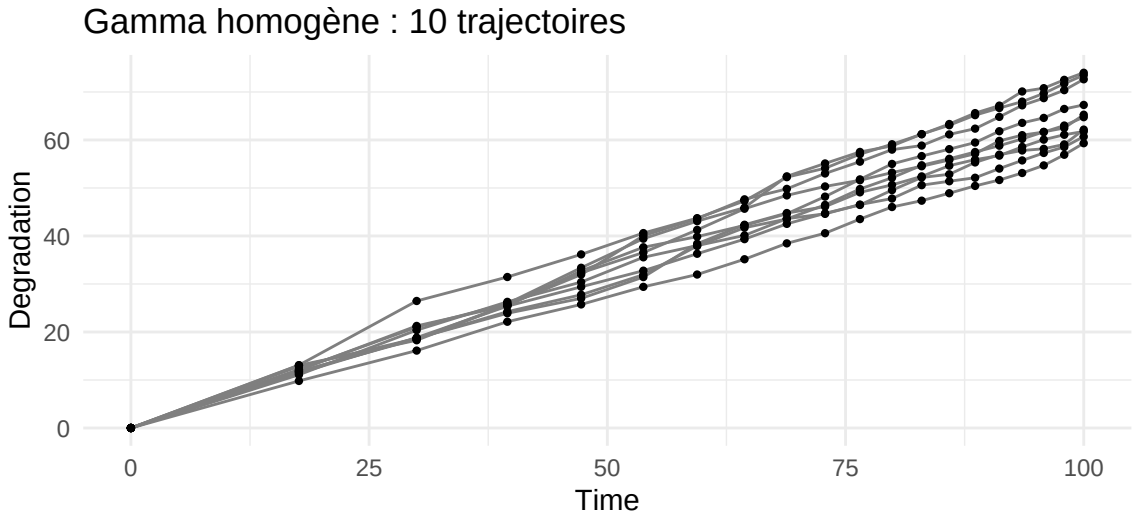


Modèle 1 : gamma

```
set.seed(121)
x <- sim_gamma(p2, a = 2, b = 3)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 17.63450 11.09319 13.09138 12.17468
## [3,] 29.98974 20.37922 26.45837 21.00282
## [4,] 39.50887 26.27595 31.45178 25.91770
## [5,] 47.25388 30.39401 36.16545 33.37539
## [6,] 53.78331 35.55453 40.60457 39.43310
```

```
plot_traj(x, "Gamma homogène : 10 trajectoires")
```

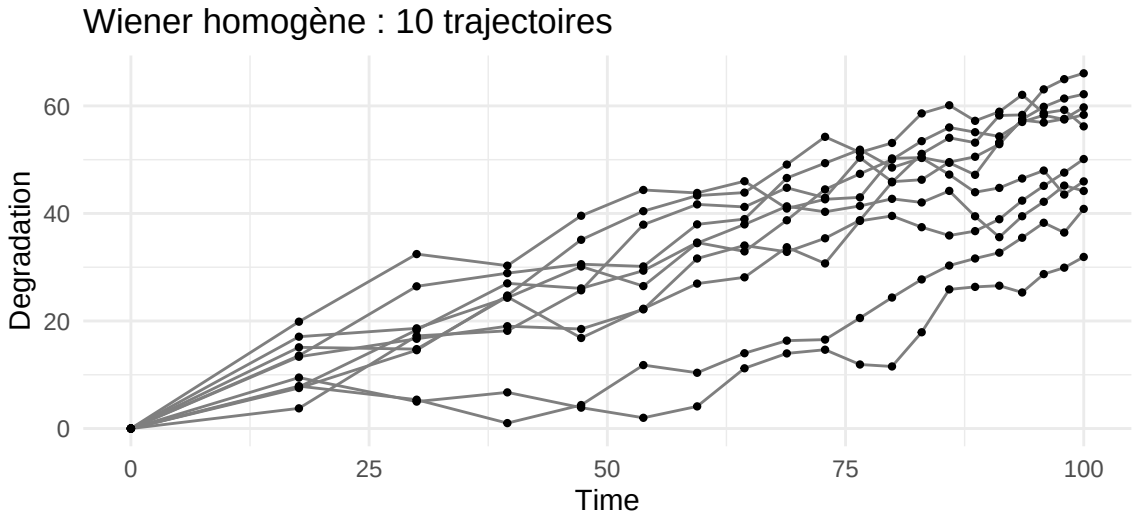


Modèle 2 : Wiener

```
set.seed(122)
x <- sim_wiener(p2, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000 0.000000
## [2,] 17.63450 17.07338 13.34670 9.472999
## [3,] 29.98974 18.63306 16.68864 5.008735
## [4,] 39.50887 24.31601 19.02940 6.723054
## [5,] 47.25388 30.13363 18.50651 3.880899
## [6,] 53.78331 26.49122 22.16033 1.993831
```

```
plot_traj(x, "Wiener homogène : 10 trajectoires")
```

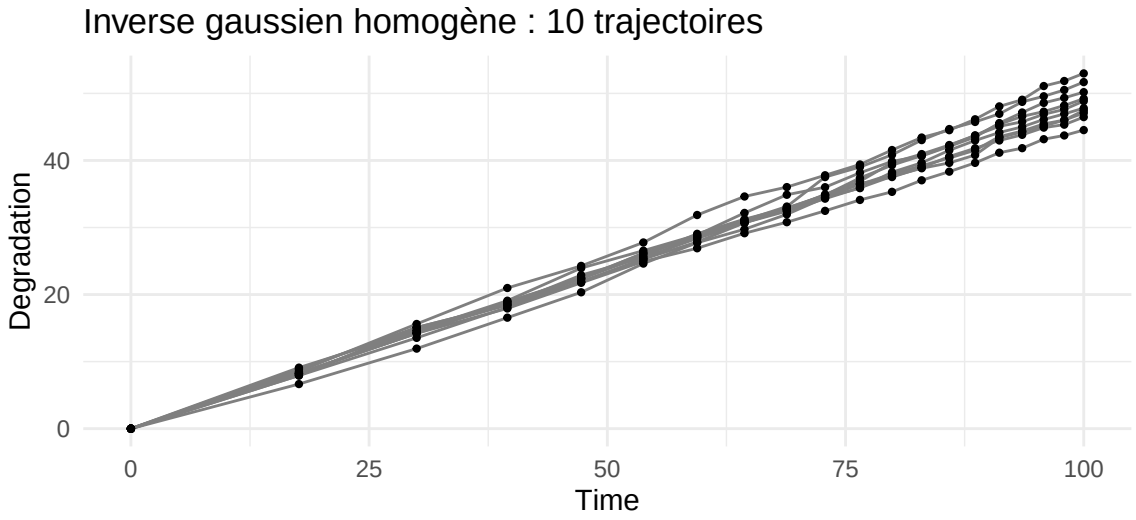


Modèle 3 : inverse gaussien

```
set.seed(123)
x <- sim_ig(p2, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##             t        X1        X2        X3
## [1,]  0.00000  0.000000  0.000000  0.000000
## [2,] 17.63450  8.248148  8.126903  8.224491
## [3,] 29.98974 14.226786 14.592250 14.294230
## [4,] 39.50887 17.926372 18.442009 18.890120
## [5,] 47.25388 21.750129 21.919203 22.507345
## [6,] 53.78331 24.933294 25.267645 26.109115
```

```
plot_traj(x, "Inverse gaussien homogène : 10 trajectoires")
```



Modèle 4 : variance-gamma

```
set.seed(124)
x <- sim_vg(p2, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##             t       X1        X2         X3
## [1,]  0.00000 0.000000  0.000000  0.0000000
## [2,] 17.63450 1.419466  9.911696 -0.2668853
## [3,] 29.98974 2.373640 15.029232  4.2291881
## [4,] 39.50887 6.909912 22.816187  2.6387083
## [5,] 47.25388 4.586146 26.025171  0.4401348
## [6,] 53.78331 9.492108 27.054773  7.0495887
```

```
plot_traj(x, "Variance-gamma homogène : 10 trajectoires")
```

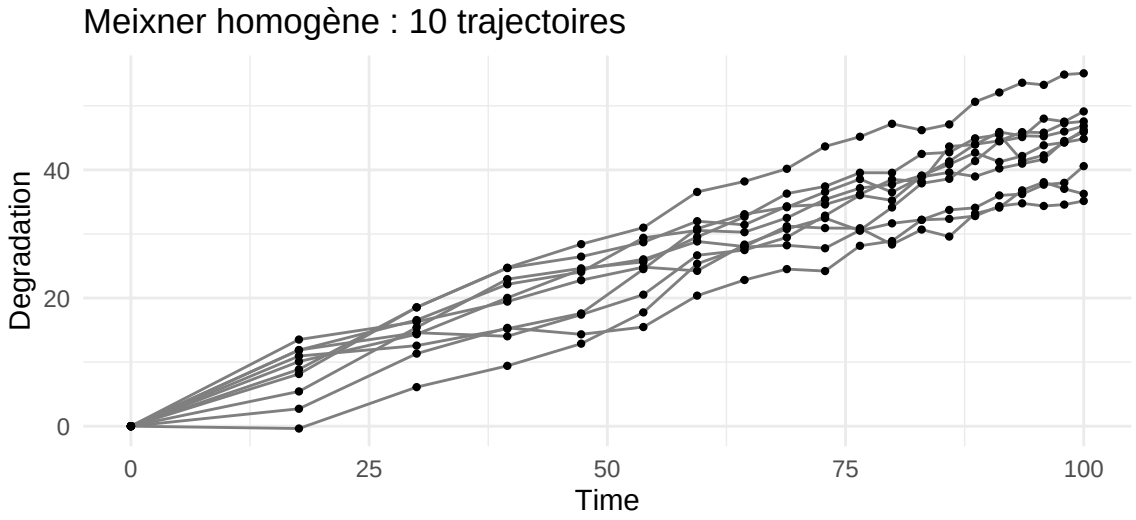


Modèle 5 : Meixner

```
set.seed(125)
x <- sim_meixner(p2, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

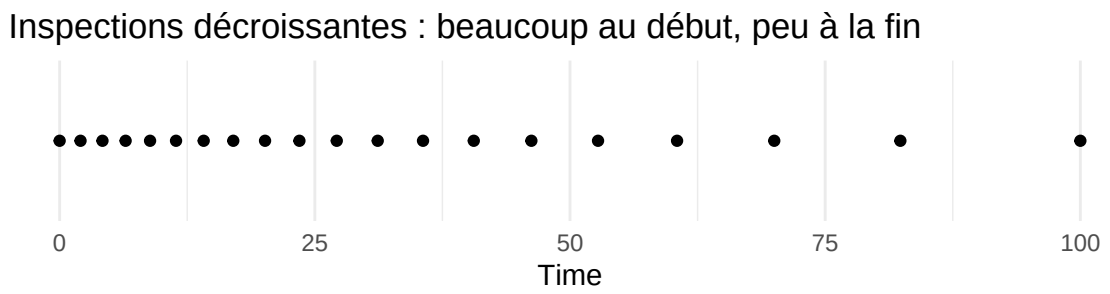
```
##             t       X1       X2       X3
## [1,]  0.00000  0.00000  0.00000  0.00000
## [2,] 17.63450 10.94291 10.08426 11.87138
## [3,] 29.98974 12.57221 14.35015 16.56729
## [4,] 39.50887 15.20837 20.02635 22.14132
## [5,] 47.25388 17.60455 24.46681 24.07085
## [6,] 53.78331 24.51319 26.04517 29.39927
```

```
plot_traj(x, "Meixner homogène : 10 trajectoires")
```



Cas 3 : inspections décroissantes

```
p3 <- inspection_times("decreasing", n = 19, n_traj = 10, tmax = 100)
plot_plan(p3$time[p3$id == 1], "Inspections décroissantes : beaucoup au début, peu à la fin")
```

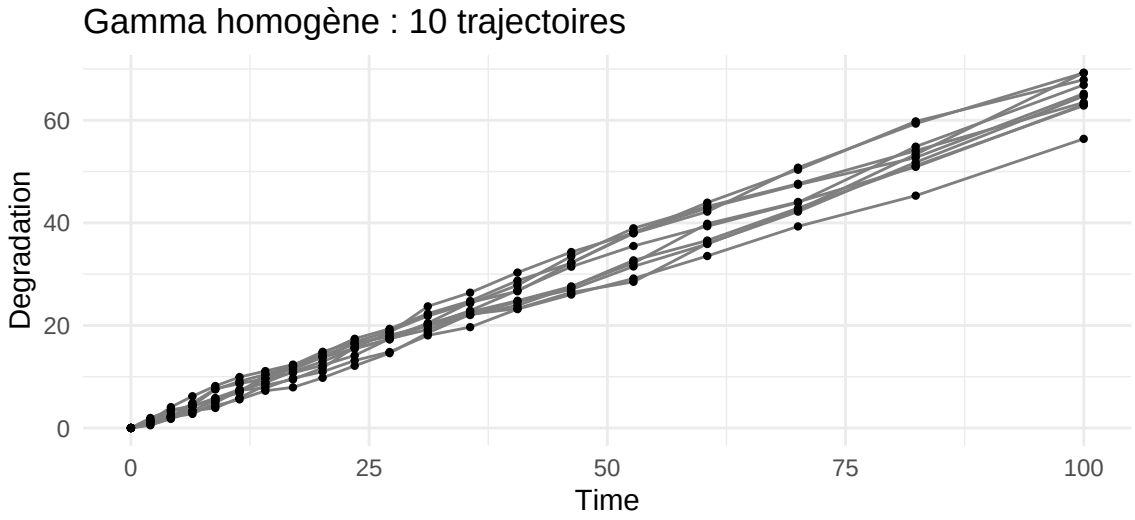


Modèle 1 : gamma

```
set.seed(131)
x <- sim_gamma(p3, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.044686 0.7352141 1.345022 0.5348972
## [3,]  4.194814 1.8045992 2.541255 1.8324493
## [4,]  6.461852 3.1807416 3.395325 4.3396604
## [5,]  8.859250 4.3182935 3.906419 7.5080730
## [6,] 11.402919 5.5752760 5.900023 9.0922130
```

```
plot_traj(x, "Gamma homogène : 10 trajectoires")
```

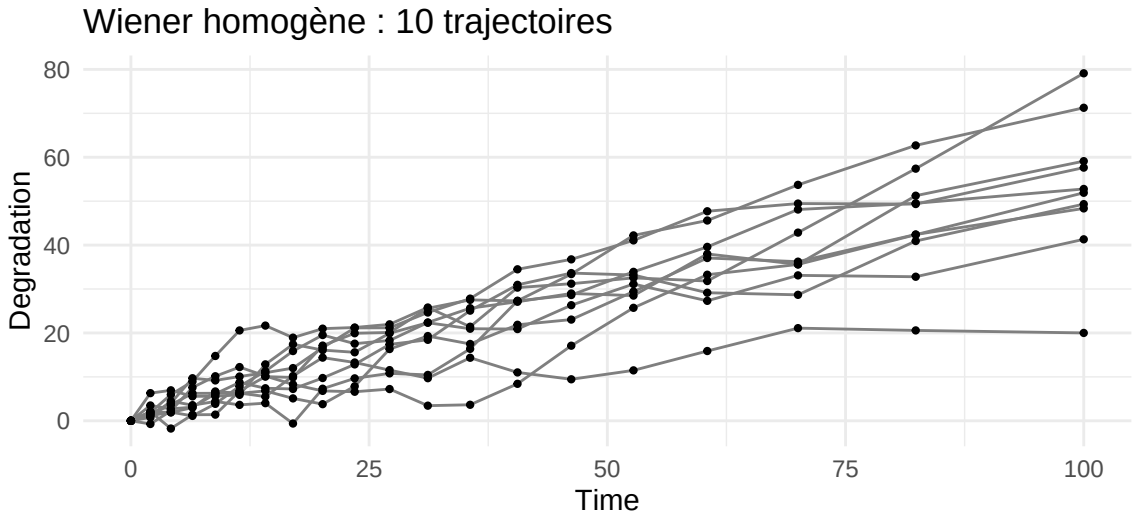


Modèle 2 : Wiener

```
set.seed(132)
x <- sim_wiener(p3, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t       X1       X2        X3
## [1,]  0.000000 0.000000 0.000000  0.000000
## [2,]  2.044686 2.039259 3.487764  1.565567
## [3,]  4.194814 1.893723 2.714529  4.031155
## [4,]  6.461852 3.004754 3.097378  9.687832
## [5,]  8.859250 6.681576 6.134371  9.187424
## [6,] 11.402919 6.249524 8.737552 10.076563
```

```
plot_traj(x, "Wiener homogène : 10 trajectoires")
```

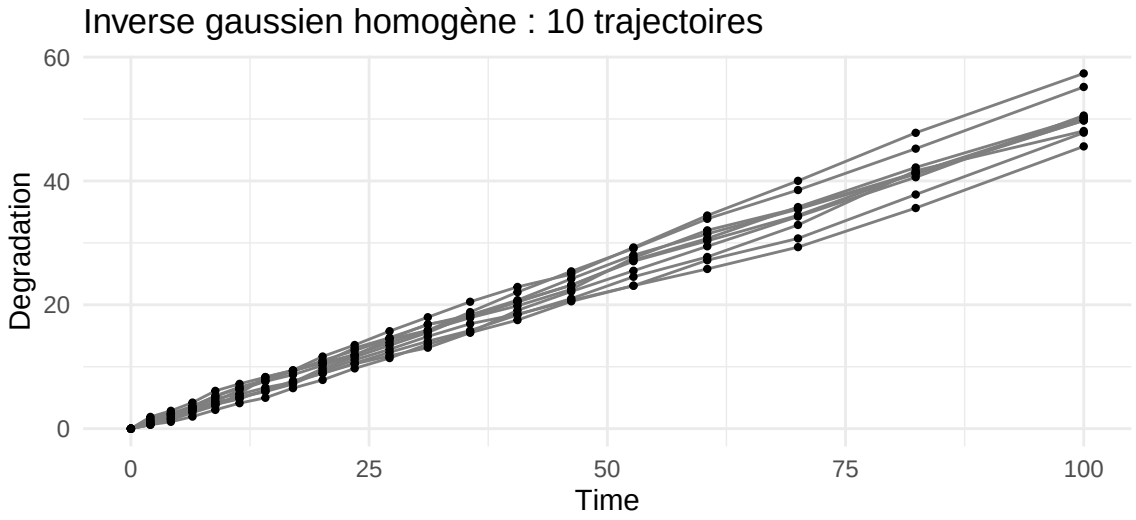


Modèle 3 : inverse gaussien

```
set.seed(133)
x <- sim_ig(p3, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000 0.0000000 0.000000 0.0000000
## [2,]  2.044686 0.9905369 1.247812 0.7661222
## [3,]  4.194814 2.2006585 2.255756 1.5841530
## [4,]  6.461852 3.0720847 3.668087 3.5023778
## [5,]  8.859250 4.3903777 5.198115 4.8321986
## [6,] 11.402919 6.2607374 6.557040 5.6310647
```

```
plot_traj(x, "Inverse gaussien homogène : 10 trajectoires")
```

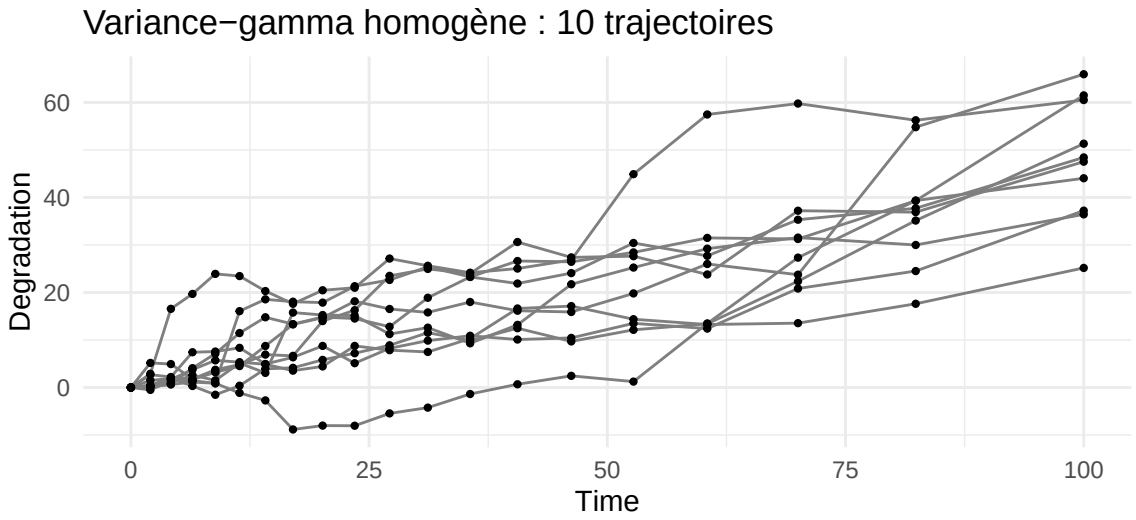


Modèle 4 : variance-gamma

```
set.seed(134)
x <- sim_vg(p3, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1        X2         X3
## [1,]  0.000000 0.0000000  0.000000  0.0000000
## [2,]  2.044686 0.1856678  2.895859  0.1090180
## [3,]  4.194814 1.6761704 16.557590  0.6399658
## [4,]  6.461852 3.7614427 19.695813  1.1370570
## [5,]  8.859250 5.7122171 23.913441  0.8924149
## [6,] 11.402919 5.3310732 23.453153 16.0690973
```

```
plot_traj(x, "Variance-gamma homogène : 10 trajectoires")
```

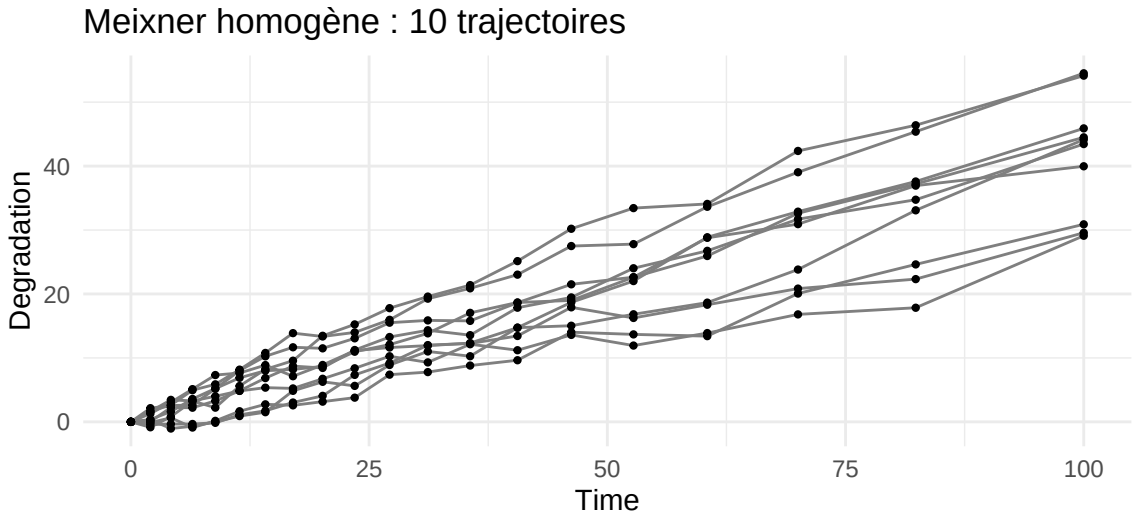


Modèle 5 : Meixner

```
set.seed(135)
x <- sim_meixner(p3, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t          X1        X2       X3
## [1,]  0.000000  0.00000000 0.0000000 0.000000
## [2,]  2.044686  0.31631140 0.0555588 1.291301
## [3,]  4.194814 -1.04229570 2.2365168 3.047314
## [4,]  6.461852 -0.68799770 2.2140710 5.117104
## [5,]  8.859250 -0.01440197 3.2786644 7.320675
## [6,] 11.402919  0.90549898 4.8513695 7.637460
```

```
plot_traj(x, "Meixner homogène : 10 trajectoires")
```



Cas 4 : mixtes, beaucoup au début et à la fin

```
p4 <- inspection_times("dec-inc", n = 19, n_traj = 10, tmax = 100)
plot_plan(p4$time[p4$id == 1], "Beaucoup au début et à la fin, peu au milieu")
```

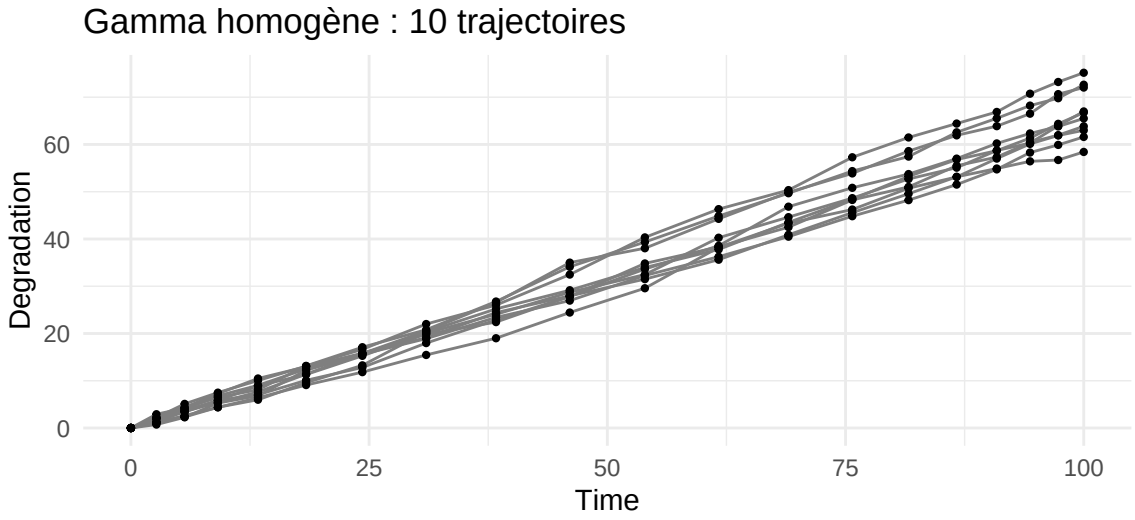


Modèle 1 : gamma

```
set.seed(141)
x <- sim_gamma(p4, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2         X3
## [1,]  0.000000  0.000000  0.000000  0.0000000
## [2,]  2.679282  2.019488  1.957203  0.7280362
## [3,]  5.638566  3.739231  3.377459  2.2594259
## [4,]  9.127515  5.788542  5.847097  5.2725973
## [5,] 13.338389  8.920814  7.039203  6.8201854
## [6,] 18.385222 11.314482 10.082758 12.2277362
```

```
plot_traj(x, "Gamma homogène : 10 trajectoires")
```

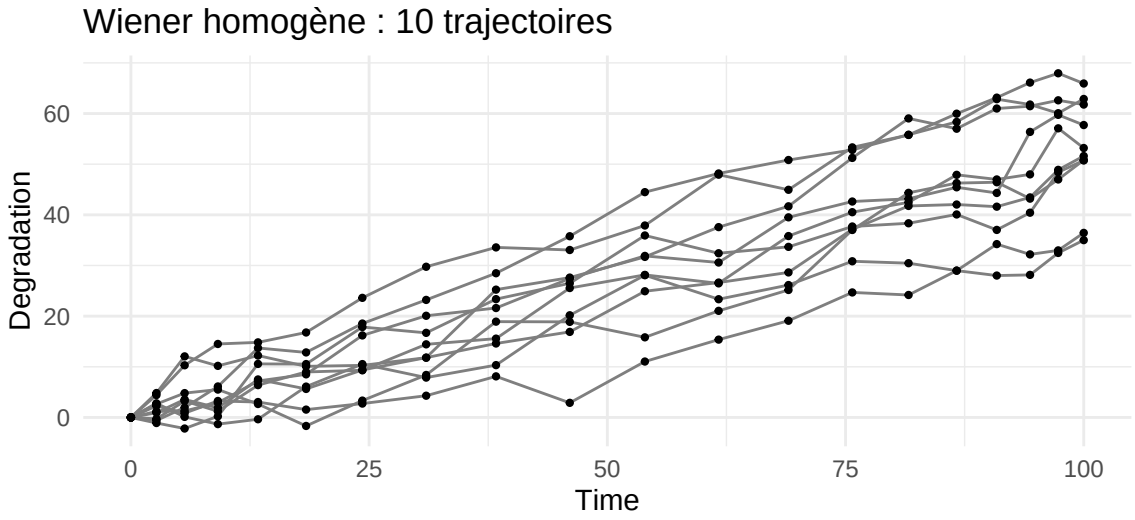


Modèle 2 : Wiener

```
set.seed(142)
x <- sim_wiener(p4, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t        X1         X2         X3
## [1,]  0.000000  0.000000  0.0000000  0.0000000
## [2,]  2.679282  4.420328  2.7880040 -1.0967920
## [3,]  5.638566 10.297168  0.1124737 -2.1863975
## [4,]  9.127515 14.508139 -1.3165312  0.2204537
## [5,] 13.338389 14.823545 -0.3581420 10.5689549
## [6,] 18.385222 16.776834  6.0893315 10.5384770
```

```
plot_traj(x, "Wiener homogène : 10 trajectoires")
```

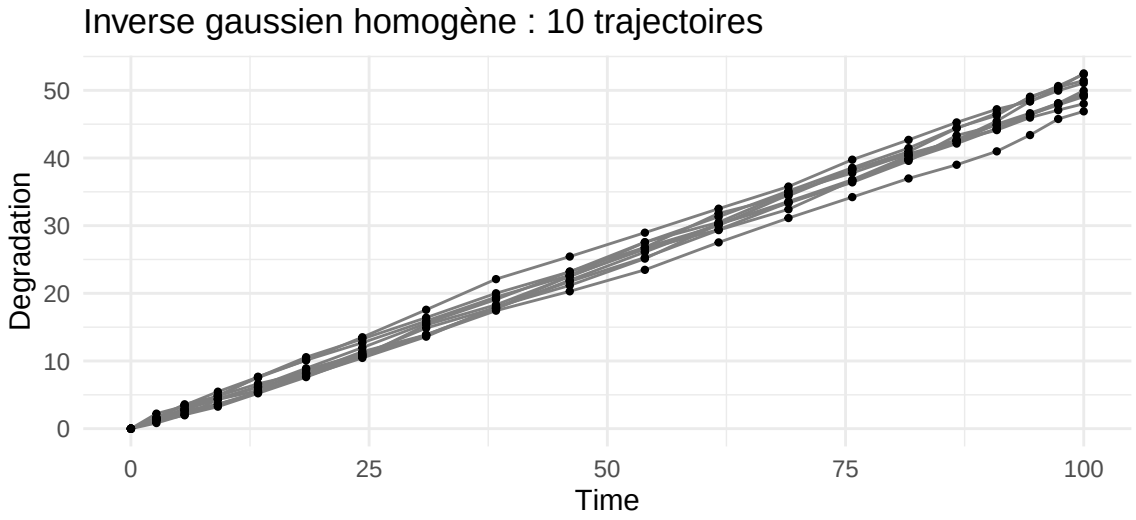


Modèle 3 : inverse gaussien

```
set.seed(143)
x <- sim_ig(p4, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000 0.0000000  0.000000  0.000000
## [2,]  2.679282 0.8299491  1.622756  1.326539
## [3,]  5.638566 2.0564501  3.603001  2.677501
## [4,]  9.127515 3.2484546  4.775965  4.910051
## [5,] 13.338389 5.2230838  7.672906  7.548331
## [6,] 18.385222 7.6192727 10.108601 10.121467
```

```
plot_traj(x, "Inverse gaussien homogène : 10 trajectoires")
```



Modèle 4 : variance-gamma

```
set.seed(144)
x <- sim_vg(p4, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1       X2         X3
## [1,]  0.000000  0.000000  0.00000  0.0000000
## [2,]  2.679282  9.077875 11.86655  0.3547826
## [3,]  5.638566 10.424350 13.37995  3.3131547
## [4,]  9.127515 10.455647 13.16475  2.9083206
## [5,] 13.338389 12.685364 18.34240 -0.1231431
## [6,] 18.385222 11.128890 21.02556  4.0621666
```

```
plot_traj(x, "Variance-gamma homogène : 10 trajectoires")
```

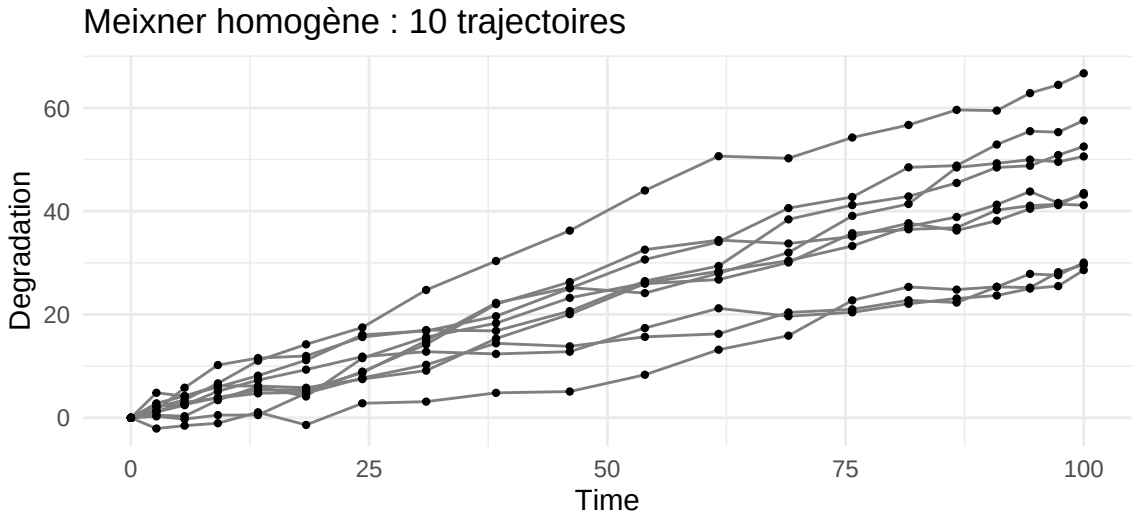


Modèle 5 : Meixner

```
set.seed(145)
x <- sim_meixner(p4, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

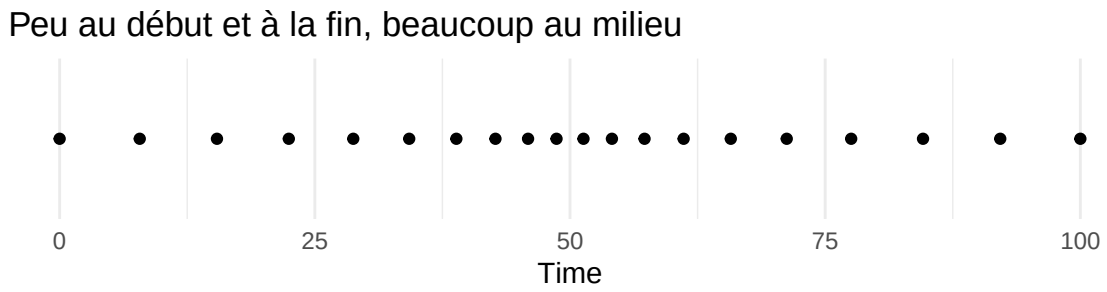
```
##              t        X1         X2       X3
## [1,]  0.000000  0.000000  0.0000000 0.000000
## [2,]  2.679282  2.090297  0.2763308 4.837432
## [3,]  5.638566  3.453719 -0.2519603 4.196843
## [4,]  9.127515  6.665677  0.5233437 6.211717
## [5,] 13.338389 11.044790  0.5365556 6.157052
## [6,] 18.385222 14.221212  4.7681305 5.814885
```

```
plot_traj(x, "Meixner homogène : 10 trajectoires")
```



Cas 5 : mixtes, beaucoup au milieu

```
p5 <- inspection_times("inc-dec", n = 19, n_traj = 10, tmax = 100)
plot_plan(p5$time[p5$id == 1], "Peu au début et à la fin, beaucoup au milieu")
```

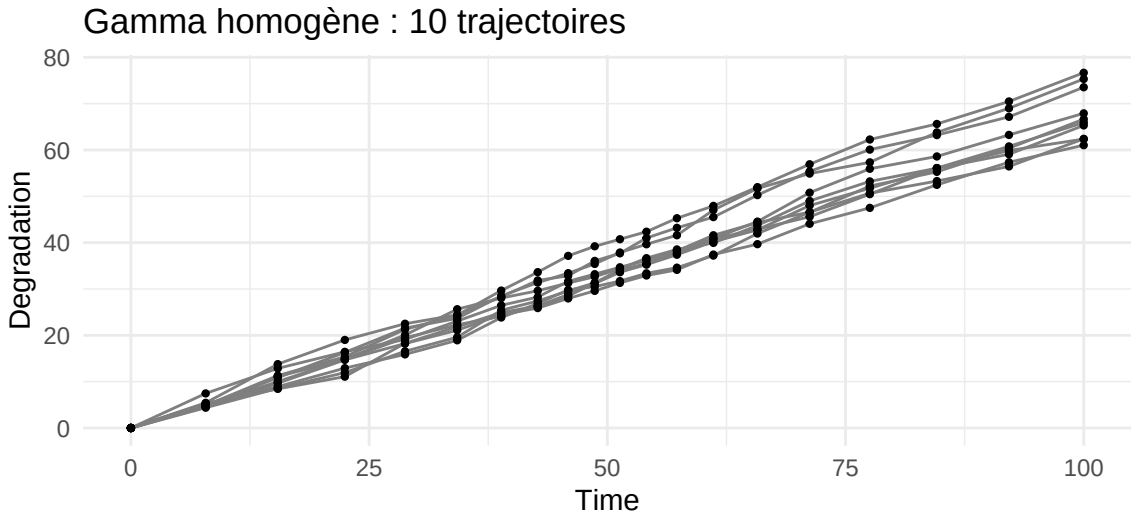


Modèle 1 : gamma

```
set.seed(151)
x <- sim_gamma(p5, a = 2, b = 3)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.847034  4.997945  5.427388  4.508088
## [3,] 15.414065 10.000425 13.784730  9.666280
## [4,] 22.451433 14.887507 19.004520 14.705898
## [5,] 28.766874 20.068298 22.497879 18.221144
## [6,] 34.246357 25.636675 24.422123 21.106650
```

```
plot_traj(x, "Gamma homogène : 10 trajectoires")
```

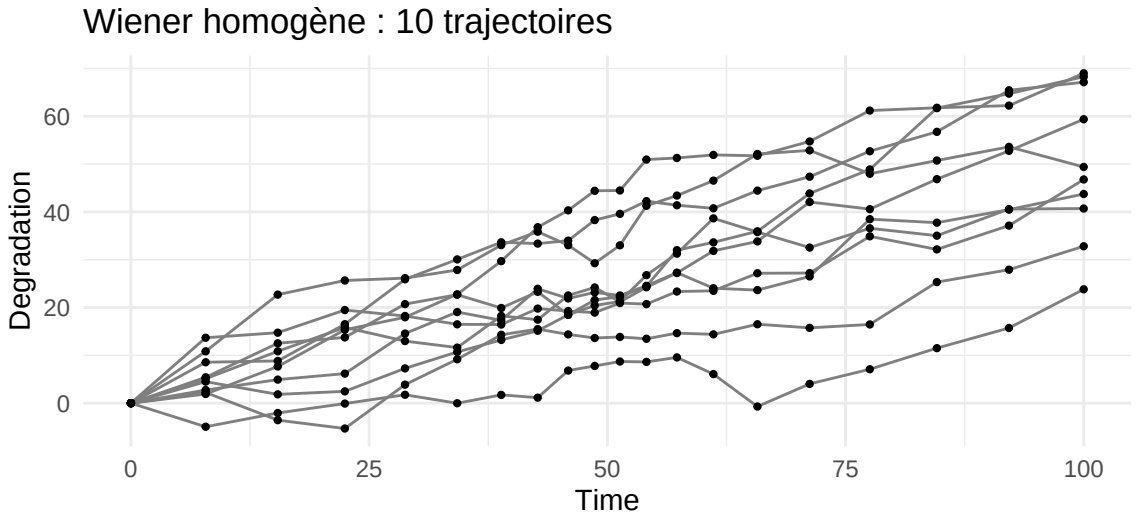


Modèle 2 : Wiener

```
set.seed(152)
x <- sim_wiener(p5, mu = 0.5, sigma = 1.5)
head(x[, 1:4])
```

```
##              t       X1        X2        X3
## [1,]  0.000000  0.00000  0.000000  0.000000
## [2,]  7.847034 13.67857  5.409302  2.218293
## [3,] 15.414065 14.73747 12.520859 -3.556313
## [4,] 22.451433 19.47012 13.747583 -5.304637
## [5,] 28.766874 18.22984 20.709516  3.867991
## [6,] 34.246357 16.48403 22.677161  9.174728
```

```
plot_traj(x, "Wiener homogène : 10 trajectoires")
```

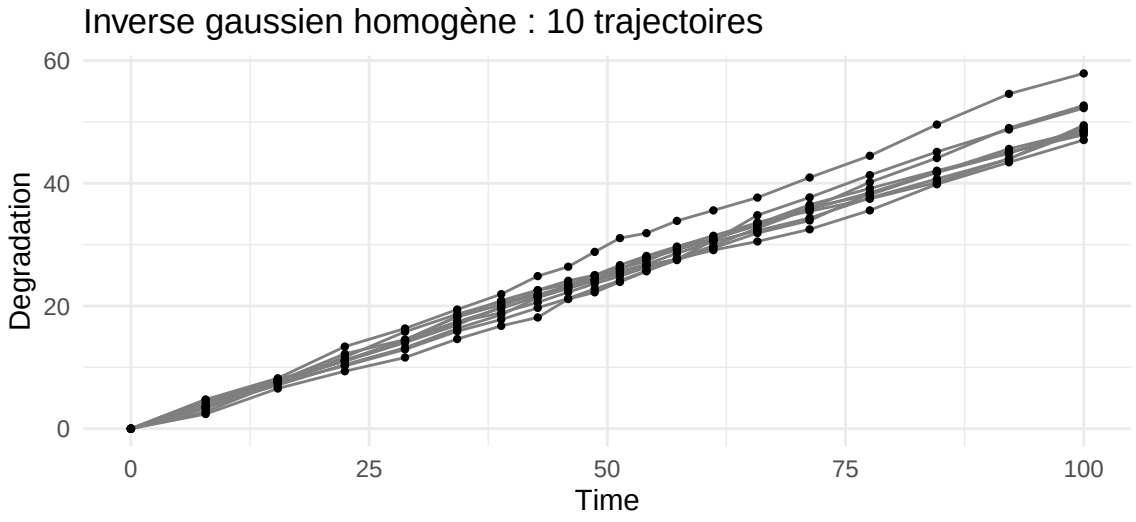


Modèle 3 : inverse gaussien

```
set.seed(153)
x <- sim_ig(p5, mu = 0.5, eta = 2)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.000000  0.000000  0.000000  0.000000
## [2,]  7.847034  2.384445  4.692209  3.441015
## [3,] 15.414065  6.546014  7.482063  7.628125
## [4,] 22.451433  9.356540 11.657376 11.096211
## [5,] 28.766874 11.593604 15.796735 14.419819
## [6,] 34.246357 14.609611 18.657331 18.303713
```

```r
plot_traj(x, "Inverse gaussien homogène : 10 trajectoires")
```

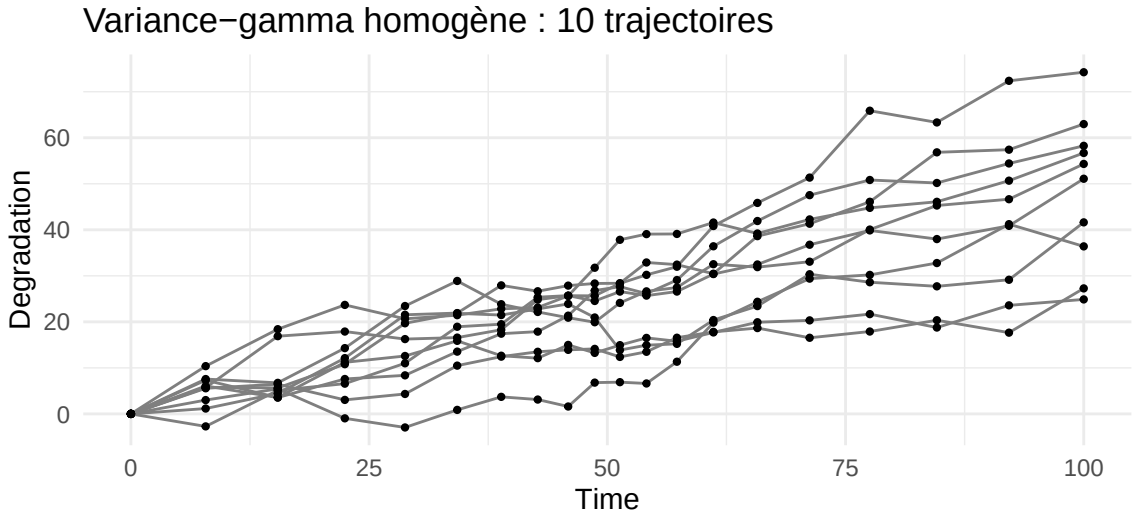


Modèle 4 : variance-gamma

```r
set.seed(154)
x <- sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000  0.00000  0.000000
## [2,]  7.847034  5.779344 10.37584  7.568184
## [3,] 15.414065  5.639402 18.40278  6.745224
## [4,] 22.451433 10.796474 23.67842 14.289071
## [5,] 28.766874 19.688026 20.65997 23.435589
## [6,] 34.246357 21.839781 21.44797 28.884128
```

```r
plot_traj(x, "Variance-gamma homogène : 10 trajectoires")
```

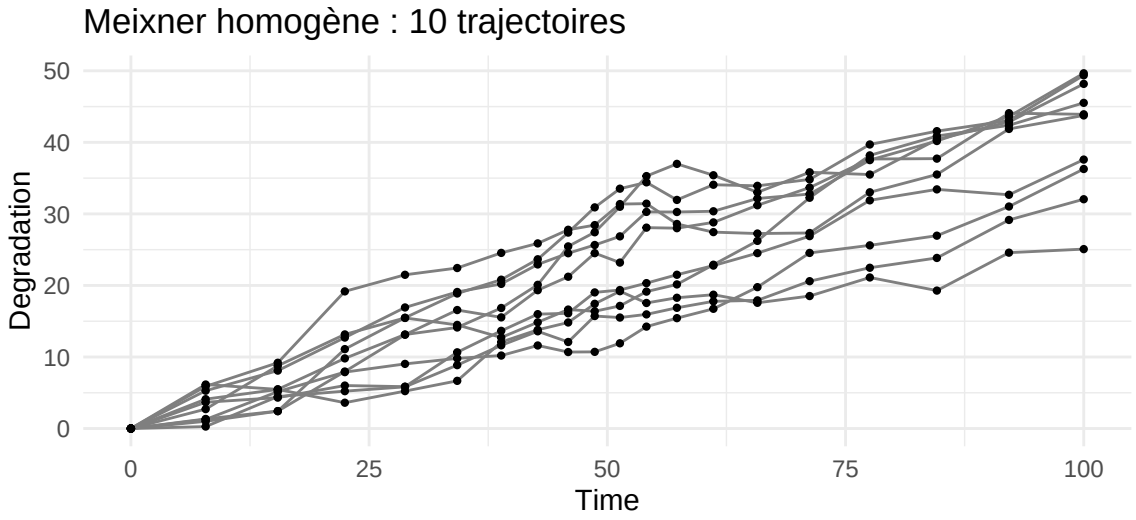


Modèle 5 : Meixner

```
set.seed(155)
x <- sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2)
head(x[, 1:4])
```

```
##              t        X1       X2        X3
## [1,]  0.000000  0.000000 0.000000  0.000000
## [2,]  7.847034  5.286274 1.263592  5.887246
## [3,] 15.414065  8.093605 5.154572  9.193110
## [4,] 22.451433 12.712238 7.883720 19.169830
## [5,] 28.766874 16.922962 9.032652 21.488630
## [6,] 34.246357 19.086135 9.805271 22.436828
```

```
plot_traj(x, "Meixner homogène : 10 trajectoires")
```



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III - Processus de Lévy non homogènes

Un changement de temps A (fonction croissante, continue à droite avec des limites à gauche, $A(0) = 0$ et $A(t) \to +\infty$) donne le processus $X_t = Y_{A(t)}$, où Y est un processus de Lévy. Le paramètre que A remplace (identifiabilité) n'est alors pas donné :

Processus	Fonction	$X_t - X_s$, $\Delta A = A(t) - A(s)$	Avec $A(t) = at$ ou μt
Gamma	`sim_gamma(t, b, A)`	$\Gamma(\Delta A, b)$	gamma homogène de forme a
Wiener	`sim_wiener(t, sigma, A)`	$\mathcal{N}(\Delta A, \sigma^2 \Delta A)$	Wiener homogène $(\mu, \sigma\sqrt{\mu})$
Inverse gaussien	`sim_ig(t, eta, A)`	$IG(\Delta A, \eta \Delta A^2)$	IG homogène $(\mu, \eta\mu^2)$

Pour le Wiener, $X_t = \mu t + \sigma B_{\mu t}$ a la loi de $\mu t + \sigma\sqrt{\mu}\, B_t$, car $(B_{\mu t})_{t\geq 0}$ a la loi de $(\sqrt{\mu}\, B_t)_{t\geq 0}$.

On prend $A(t) = 0.02\, t^{1.5}$, de sorte que $A(100) = 20$, et les 10 trajectoires du cas 2 de la partie II (inspections

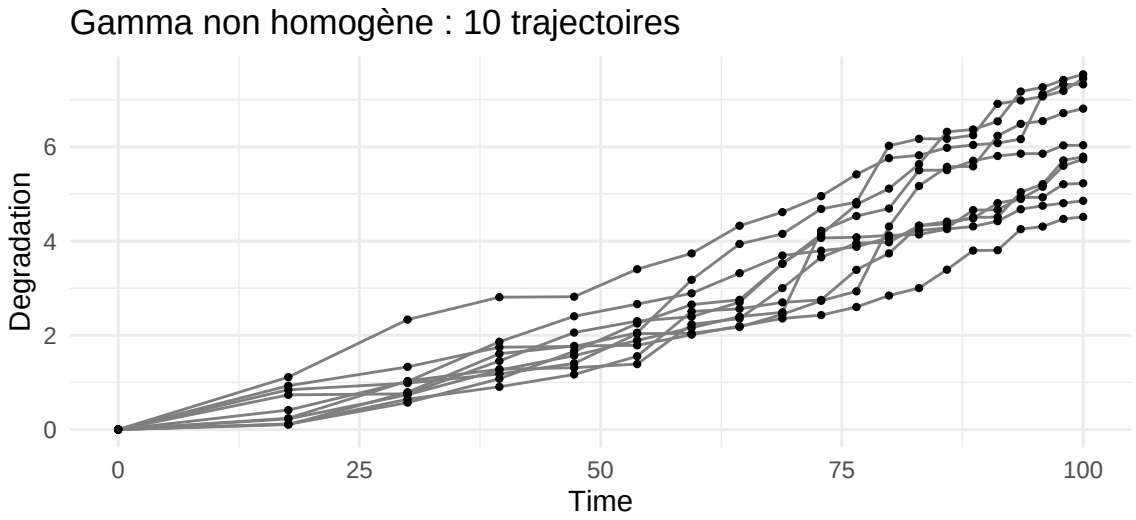
croissantes).

Modèle 1 : gamma non homogène (Cha et Mercier, 2019, exemple 2)

```
A <- function(t) 0.02 * t^1.5
set.seed(301)
x <- sim_gamma(p2, b = 3, A = A)
x
```

```
## 10 simulated trajectory(ies) of the non-homogeneous gamma process, same inspection times (matrix)
## parameters: b = 3
## time change: A = function (t)  0.02 * t^1.5
##
##              t       X1        X2        X3        X4       X5
## [1,]  0.00000 0.000000 0.0000000 0.0000000 0.0000000 0.000000
## [2,] 17.63450 0.413235 0.2260551 0.1117200 0.1195689 1.113368
## [3,] 29.98974 1.018565 0.7377684 0.5715126 0.6429133 2.333316
## [4,] 39.50887 1.863933 1.2619976 1.0769566 0.9074442 2.809108
## [5,] 47.25388 2.404166 1.5691255 1.6593942 1.1687358 2.819350
## [6,] 53.78331 2.662670 1.8916061 2.2509750 1.5558784 3.403399
```

```
plot_traj(x, "Gamma non homogène : 10 trajectoires")
```



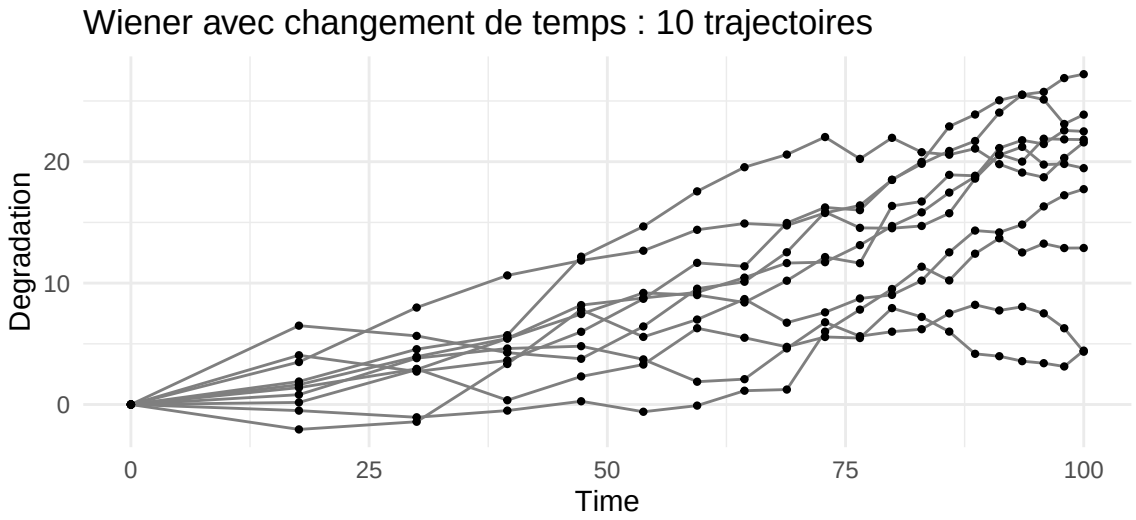
Modèle 2 : Wiener avec changement de temps (Cha et Mercier, 2019, exemple 1)

```
set.seed(302)
x <- sim_wiener(p2, sigma = 1.5, A = A)
head(x[, 1:4])
```

```
##              t        X1        X2        X3
## [1,]  0.00000  0.000000 0.0000000 0.0000000
## [2,] 17.63450  1.893327 0.1725442 0.8115244
## [3,] 29.98974  4.555056 2.8800225 3.8168637
## [4,] 39.50887  5.712354 5.4312405 4.6137487
```

```
## [5,] 47.25388 12.176191 7.4521673 4.8047256
## [6,] 53.78331 14.665366 9.1931509 3.7176109
```

```
plot_traj(x, "Wiener avec changement de temps : 10 trajectoires")
```

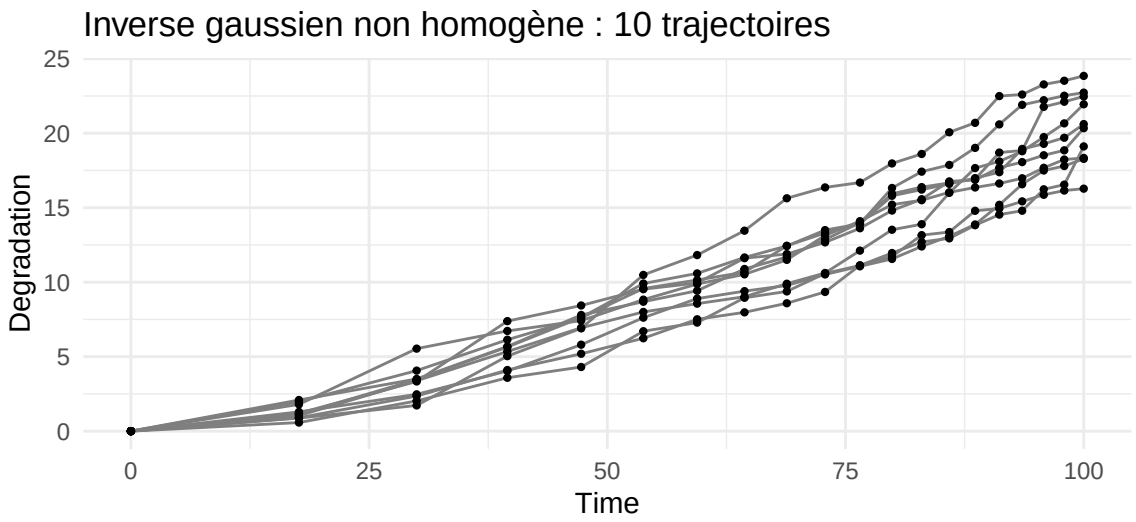


Modèle 3 : inverse gaussien non homogène (Cha et Mercier, 2019, exemple 3)

```
set.seed(303)
x <- sim_ig(p2, eta = 2, A = A)
head(x[, 1:4])
```

```
##             t       X1       X2         X3
## [1,]  0.00000 0.000000 0.000000  0.0000000
## [2,] 17.63450 1.974875 1.205460  0.8787256
## [3,] 29.98974 4.063401 3.334509  1.7250852
## [4,] 39.50887 6.136971 7.385009  5.0342163
## [5,] 47.25388 7.608008 8.435758  6.9013147
## [6,] 53.78331 9.573188 9.529091 10.4882354
```

```
plot_traj(x, "Inverse gaussien non homogène : 10 trajectoires")
```



Vérification : $A(t) = at$ redonne le processus homogène

Avec la même graine, le gamma homogène $(a, b) = (2, 3)$ et le gamma non homogène avec $A(t) = 2t$ donnent les mêmes trajectoires :

```
set.seed(304)
y1 <- sim_gamma(p2, a = 2, b = 3)
set.seed(304)
y2 <- sim_gamma(p2, b = 3, A = function(t) 2 * t)
all.equal(unclass(y1)[, -1], unclass(y2)[, -1], check.attributes = FALSE)
```

```
## [1] TRUE
```